

Unit 7 Overview: Area

Unit Narrative



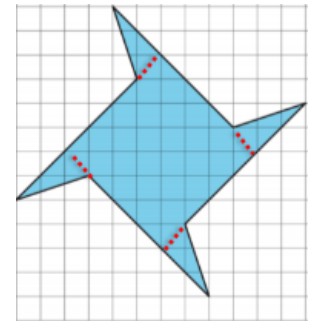
In this unit, students build on their prior knowledge and skills from Grade 5 in determining the area and perimeter of rectangles to derive the formula for the area of a triangle. Students then learn to find the areas of special quadrilaterals, composite figures, and polygons by decomposing figures into rectangles and triangles. Throughout the unit, students connect new concepts with prior knowledge as they discover the formulas for finding area of various figures and then use them to solve problems.

Section 1 begins with establishing the relationship between the area of triangles and the area of rectangles. Students explore this relationship to derive a formula to find the area of triangle. Instruction includes determining the base and height of any triangle and using these dimensions to find the area.

Section 2 continues the focus on area, into finding the area of quadrilaterals other than rectangles. Students first learn to decompose quadrilaterals into rectangles and triangles to find the area. Then, students discover the formulas for finding the areas of trapezoids, parallelograms, and rhombi. Students use these formulas to solve mathematical and real-world problems.

In Section 3, students practice various approaches to finding area of composite shapes, including decomposing into quadrilaterals and triangles before finding the total area of all the decomposed figures. This section also includes finding the missing dimensions of various two-dimensional and composite figures. Students decompose the areas of polygons and composite figures into areas of rectangles and triangles to determine the area.

This unit builds a solid foundation of area before moving into Unit 8 where students will be introduced to the concept of surface area using nets.



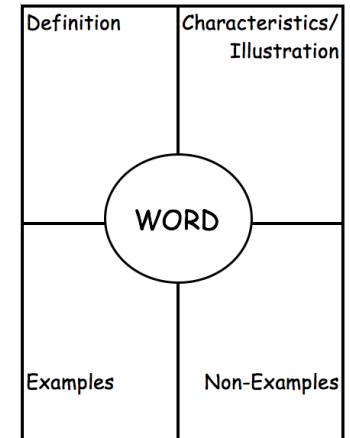
Unit At-A-Glance

Section 1: Exploring Area of Triangles	Benchmarks Addressed	Lesson 1	Area of Triangles – Part 1
	MA.6.GR.2.1	Lesson 2	Area of Triangles – Part 2
Section 2: Decomposing Quadrilaterals to Find Area	Benchmarks Addressed	Lesson 3	Decomposing Quadrilaterals to Find Area
	MA.6.GR.2.2 MA.7.GR.1.1	Lesson 4	Deriving Area Formulas for Trapezoids, Parallelograms, and Rhombi
		Lesson 5	Finding the Area of Trapezoids, Parallelograms, and Rhombi
Section 3: Area of Composite Figures	Benchmarks Addressed	Lesson 6	Area of Composite Figures
	MA.6.GR.2.2 MA.7.GR.1.2	Lesson 7	Finding the Area of Polygons
		Lesson 8	Finding the Area of Composite Figures

ELL Information

In this unit, students will be using key terms related to area. To solidify understanding of these words in context, have students create Frayer Models for the words in this unit and have them available as a reference throughout the unit.

- A Frayer Model (similar to the four-square graphic organizer) is a helpful graphic organizer because it allows students to see terms in multiple ways. For this section, students will be using many words that connect and build upon geometric understanding of two-dimensional and three-dimensional figures.
- Provide the definition but have students work in pairs to illustrate the word (including the shape/figure and the net if applicable), list examples, and list non-examples.
- For examples and nonexamples, encourage students to consider this as a work in progress and to add throughout the section as they are presented.
- Consider creating a Frayer Model for each term from the Glossary by Section lists (below).



All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from “Additional Differentiated Support.”

- | | |
|--------------------|-------------------|
| • Acute triangle | • Area |
| • Composite figure | • Obtuse triangle |
| • Parallelogram | • Polygon |
| • Quadrilateral | • Rectangle |
| • Regular polygon | • Rhombus |
| • Right triangle | • Trapezoid |

Additional Differentiated Support

Scaffolding Support	On-Ramp
Mark-Up Text In this unit, students will be solving real-world problems. When students translate words into numbers, algebraic expressions, or equations, encourage them to use a highlighter to highlight phrases and words to call attention to important information. Using a small piece of text, model for students how you would mark up the text before releasing them to do the same. Students will read the text slowly and carefully, focusing on one sentence at a time and marking the text based on a key. Then they will turn to a partner to discuss each sentence and compare where they made marks. For the first few examples, have groups share their decisions with the class. Then, guide students through turning the marked-up text into an expression. ELL students may find this a helpful intermediary step, so encourage them to use this strategy even as you release them to work independently.	On-Ramp to 6th Grade Math provides video and practice tools for students to scaffold the following skills: • Using the Area Formula for Rectangles • Calculating Volume • Finding Area of Rectilinear Shapes • Creating Composite Shapes • Classifying Quadrilaterals • Finding Area of Figures Made from Unit Squares



Section Overview

Section 1: Exploring Area of Triangles (Lessons 1-2)

Benchmarks Addressed	Learning Targets	
MA.6.GR.2.1 Derive a formula for the area of a right triangle using a rectangle. Apply a formula to find the area of a triangle. <i>Clarification 1: Instruction focuses on the relationship between the area of a rectangle and the area of a right triangle.</i> <i>Clarification 2: Within this benchmark, the expectation is to know from memory a formula for the area of a triangle.</i>	Lesson 1	I can discover the formula for the area of a triangle.
	Lesson 2	I can apply a formula to find the area of triangles.

Section 2: Decomposing Quadrilaterals to Find Area (Lessons 3-5)

Benchmarks Addressed	Learning Targets	
MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles. <i>Clarification 1: Problem types include finding area of composite shapes and determining missing dimensions.</i> <i>Clarification 2: Within this benchmark, the expectation is to know from memory a formula for the area of a rectangle and triangle.</i> <i>Clarification 3: Dimensions are limited to positive rational numbers.</i> MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms and rhombi. <i>Clarification 1: Instruction focuses on the connection from the areas of trapezoids, parallelograms and rhombi to the areas of rectangles or triangles.</i> <i>Clarification 1: Within this benchmark, the expectation is not to memorize area formulas for trapezoids, parallelograms and rhombi.</i>	Lesson 3	I can find the area of quadrilaterals by decomposing them into triangles and rectangles.
	Lesson 4	I can use the formulas for the area of triangles and rectangles to derive formulas for the area of trapezoids, parallelograms, and rhombi.
	Lesson 5	I can find the area of trapezoids, parallelograms, and rhombi.

Section 3: Area of Composite Figures (Lessons 6-8)

Benchmarks Addressed	Learning Targets	
MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles. - Clarification 1: Problem types include finding area of composite shapes and determining missing dimensions. - Clarification 2: Within this benchmark, the expectation is to know from memory a formula for the area of a rectangle and triangle. - Clarification 3: Dimensions are limited to positive rational numbers. MA.7.GR.1.2 Solve mathematical or real-world problems involving the area of polygons or composite figures by decomposing them into triangles or quadrilaterals. - Clarification 1: Within this benchmark, the expectation is not to find areas of figures on the coordinate plane or to find missing dimensions.	Lesson 6	- I can find the area of composite figures. - I can find the value of a missing dimension for a quadrilateral or a composite figure.
	Lesson 7	- I can decompose a regular polygon into triangles to find the area. - I can decompose a regular polygon into triangles and quadrilaterals to find the area.
	Lesson 8	I can decompose composite figures into triangles and quadrilaterals to find the area.



Vertical Alignment

Grade 6 Accelerated Unit 7
Area



Geometry Unit 9 and 13
Two- and Three-Dimensional Shapes
Shapes on a Coordinate Plane

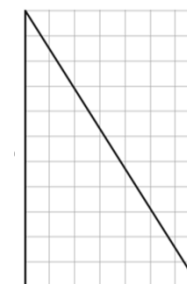
Prior Course(s)	Course Level	Future Course(s)
MA.5.GR.2.1 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas.	MA.6.GR.2.1 Derive a formula for the area of a right triangle using a rectangle. Apply a formula to find the area of a triangle. MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles. MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi. MA.7.GR.1.2 Solve mathematical or real-world problems involving the area of polygons or composite figures by decomposing them into triangles or quadrilaterals.	MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeter or area of polygons. MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.

7.1 Area of Triangles – Part 1

Lesson Introduction

 Benchmark	MA.6.GR.2.1 Derive a formula for the area of a right triangle using a rectangle. Apply a formula to find the area of a triangle.		 Learning Target	I can discover the formula for the area of a triangle.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.GR.2.1 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas.		MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi.	
 Essential Questions	- What is the formula to find the area of a triangle? - How can you use the area of a rectangle to find the area of a triangle?			
 Lesson Narrative	In the lesson, students will discover the formula for the area of a triangle by identifying how the area of a triangle relates to the area of a rectangle. During each exploration of a right triangle, acute triangle, and obtuse triangle, students will discover that the area of a triangle is always half the area of a rectangle with the same base and height. Students apply this knowledge to determine the formula to find the area of a triangle.			
 Component Summary	7.1.1 Warm-Up (~5 min.)	Open-ended response on the importance of struggling (<i>SE p. 297</i>)	7.1.2 Exploration (10-15 min.)	Use the area of a rectangle to discover how to find the area of a triangle (<i>SE p. 298</i>)
	7.1.3 Collaboration (20-25 min.)	Use the area of a rectangle to find the area of an obtuse and acute triangle (<i>SE p. 299</i>)	7.1.4 Bring It Together (5 min.)	Students synthesize information connecting area of a rectangle to a square (<i>SE p. 301</i>)
	7.1.5 Wrap-Up (~5 min.)	Students analyze their own understanding of how to find the area of a triangle (<i>SE p. 302</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Base - Height - Acute - Obtuse		- Importance of Struggle video - Paper triangles - Scissors - Glue (or tape) (Optional) Graph paper	7.1.1 Warm-Up contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions • In finding the height of a triangle, students may pick a side of an acute or obtuse triangle instead of the height. • Students might forget to multiply the length and width by $\frac{1}{2}$ to find the area of a triangle. • Students may forget to find the height of an obtuse triangle by extending the line. • In acute triangles, students may incorrectly try to extend the base to the dashed height line.	Things to Consider • The 7.1.2 Exploration is where students will be able to visualize how the formula for area of a triangle works and relates to a rectangle. • The lesson is a discovery of the formula for the area of a triangle. The formula is revealed in the lesson. Allow students to work through the discovery to determine the formula. • The visual in 7.1.3 Collaboration question 5 will help show students how the area of parallelograms can also relate to the area of a triangle and what to do with triangles that do not have right angles.
	Literacy and Language Routines Number Talks Consider using a Number Talks strategy discussion to visualize different ways to divide a rectangle. Draw a rectangle on the board. Ask students to come up with different ways to divide the rectangle in half. Have students share strategies until someone splits the rectangle diagonally.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations Throughout this lesson, students have the opportunity to make connections between the areas of various shapes, such as rectangles and parallelograms, and the area of triangles. Encourage students to express these connections through discussion as each activity allows.
	 Differentiate Instruction	Consider opportunities for students to engage in physical or digital manipulatives that offer kinesthetic entry points for making sense of the formula for the area of a triangle through the lens of a rectangle. • Coordinate Plane Cutouts: Students can compose and/or decompose triangles and rectangles using mini coordinate planes to highlight the relationship between their area formulas. • Tangram Triangles: Similarly, students can engage with the digital tangram triangles that target the relationship between rectangles and triangles and what that means for their area.
Lesson Notes:		



7.1.1 Warm-Up: The Importance of Struggle

Component Overview: Students will watch a video on the importance of struggling. Students will then answer open-ended questions about how they have personally struggled.



7.1.1 Warm-Up: The Importance of Struggle

When learning something new, whether it is a sport, a musical instrument, or even math, you may experience failure before you experience success. The key is to keep going even when you make mistakes. Learning takes place during the time you practice and during the time you struggle.

1. Which choice would you feel less comfortable doing?
 - Answering a question and getting it wrong.
 - Saying, "I do not know," as your answer to a question.

Answers will vary.

2. Describe a time that you persevered when learning something new.

Answers will vary.

3. When you persevered in learning, how did you overcome any initial fear(s)?

Answers will vary.



Consider positioning students to discuss when they have been in a situation that contained each option to engage auditory learners and put real-life context to the options. Take the discussion further and have students turn and talk about why failing is sometimes okay. Share a time you as an adult have failed and how it made you stronger.

7.1.2 Exploration: Discovering the Area of a Right Triangle

Component Overview: In this Exploration, students discover how the area of a rectangle is connected to the area of a right triangle. Begin with having students trace and cut out triangles. They will use these triangles to create a rectangle. The questions in this section all pertain to one set of triangles. Spend time exploring the connection between right triangles and rectangles, as the following sections will give more opportunity for actual calculations.



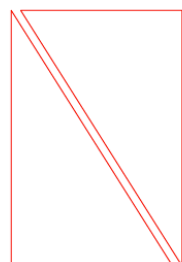
7.1.2 Exploration: Discovering the Area of a Right Triangle

A **right triangle** is shown. Each square of the grid paper is 1 centimeter (cm) by 1 cm.



- Trace two copies of the right triangle on paper, then cut them out.
- Form a rectangle using the two right triangles. Paste the triangles here.
- Using the image just created, complete the table.

	Rectangle	Triangle
Height	<u>11</u> cm	<u>11</u> cm
Base	<u>7</u> cm	<u>7</u> cm



- What is the area of the rectangle? Include units.
 77 cm^2
- Describe the relationship between the area of one right triangle and the area of the rectangle.
The area of a triangle is one-half the area of the rectangle.
- What is the formula for the area of a rectangle?
 $A = bh$
- Use the relationship observed and the formula for the area of a rectangle to write a formula that can be used to find the area of a right triangle.
 $A = \frac{1}{2}bh$



Number Talks

Consider using a Number Talks strategy discussion to visualize different ways to divide a rectangle. Draw a rectangle on the board. Ask students to come up with different ways to divide the rectangle in half. Have students share strategies until someone splits the rectangle diagonally.



Cutting and pasting will allow an entry point for kinesthetic learners and allow students to visualize how two triangles can create a rectangle.



To encourage students to make the connection between rectangles and triangles, ask questions like:

- What is the difference between a regular triangle and a right triangle?
- What does a rectangle have in common with a right triangle?
- What do you notice about the height and base of the triangle compared to the rectangle?
- What two types of measurement do you need to find the area of a triangle? What two types of measurement do you need to find the area of a rectangle? What is a different way you could write the area of a triangle?

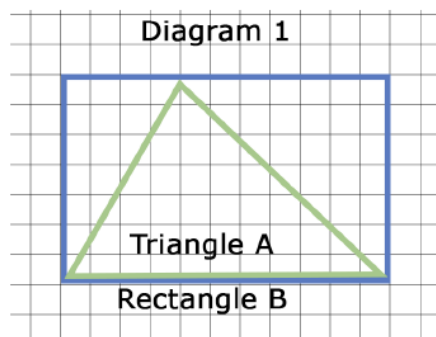
7.1.3 Collaboration: Area of an Acute and an Obtuse Triangle

Component Overview: In this Collaboration, students will work with a partner to find areas involving acute and obtuse triangles. Students then work with a partner to calculate the area of a triangle inscribed in a rectangle along with the rectangle and to find the area of obtuse triangles.

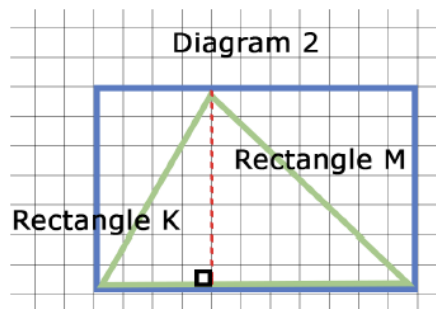


7.1.3 Collaboration: Area of an Acute and an Obtuse Triangle

An **acute triangle** is shown. Each square of the grid paper in Diagram 1 and Diagram 2 is 1 centimeter (cm) by 1 cm.



Rectangle B is drawn around acute Triangle A, as shown.



The height of acute Triangle A is shown as a dashed red line. It divides Rectangle B into two smaller rectangles, Rectangle K and Rectangle M, and divides Triangle A into two smaller right triangles.



On question 1, have students label all measurements they were given to visualize the connection as well as digest the information. Students may also need to draw the line for the height in Diagram 1 in the middle of the triangle to help visualize. This provides visual entry points into discussion through the diagrams.



Avoid student focus on the other triangles that are made outside of the triangle using the corners of the rectangle, as students may still be looking for right triangles. Discuss how a right angle should be created when finding the height of an acute or obtuse triangle.

Relating the height of Triangle A to the height of a mountain may help students understand that the height is the straight line distance from the ground (the base) to the top of the mountain (the vertex).



Consider briefly reviewing or providing definitions for acute and obtuse triangles.

7.1.3 Collaboration: Area of an Acute and an Obtuse Triangle (continued)

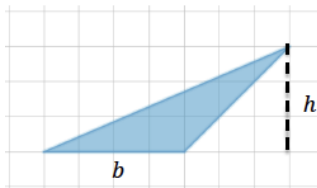
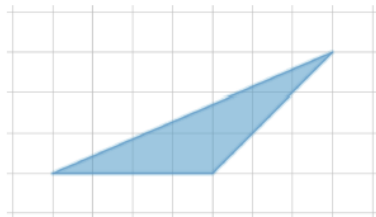
1. Use the two diagrams to complete the table.

	Area of the Rectangle	Area of the Right Triangle Drawn Inside the Rectangle
Rectangle K	28 cm^2	14 cm^2
Rectangle M	49 cm^2	24.5 cm^2

2. The area of acute Triangle A is equal to the sum of the right triangle inscribed in Rectangle K and the right triangle inscribed in Rectangle M.

- What is the area of acute Triangle A?
 38.5 cm^2
- What is the area of Rectangle B?
 77 cm^2
- What is the relationship between the area of Rectangle B and the area of acute Triangle A?
If you double the area of Triangle A, it is the same area as Rectangle B.

3. An **obtuse triangle** is shown. Each square of the grid paper is 1 centimeter (cm) by 1 cm.



- What is the length of the triangle's base (b)?
 4 cm
- What is the height (h) of the triangle?
 3 cm



7.1.3 Collaboration implements visual entry points into discussion using the diagrams. Cutting and pasting the triangles into rectangles will help kinesthetic learners relate a triangle with being half of a rectangle. Consider a Turn and Talk if some students dominate more of the conversation or if your students benefit from structured timing for conversations.



Students may struggle to find the height of triangles that are not right triangles because it may not be as clear visually. Relating the obtuse triangle to the idea of a mountain may help students visualize. Or ask students what real-world object an obtuse triangle reminds them of and use that to query about height of the triangle from the ground.

Ask prompting questions to guide students toward determining how they can find the height:

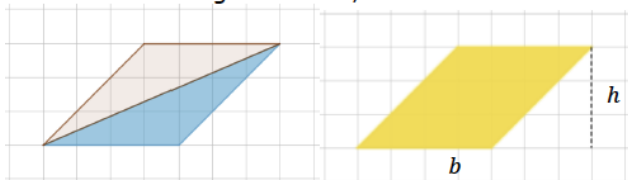
- Why isn't the longest side of the triangle the height?
- How do you know how far to draw the dotted line for the height of an obtuse triangle?



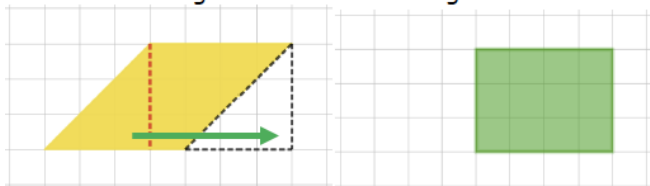
Reinforce the concept that the height needs to be perpendicular to the base. Extend the discussion by asking students to draw a different line to find the height.

7.1.3 Collaboration: Area of an Acute and an Obtuse Triangle (continued)

4. A copy of the triangle is made, and the two triangles are arranged to form a parallelogram with a base of 4 cm and a height of 3 cm, as shown.



The parallelogram is then decomposed, and the parts are rearranged to form a rectangle.



- a. Complete the table.

	Rectangle	Obtuse Triangle
Height	<u> 3 </u> cm	<u> 3 </u> cm
Base	<u> 4 </u> cm	<u> 4 </u> cm

- b. How many obtuse triangles were needed to create the rectangle?
2
- c. What is the area of the rectangle?
 12 cm^2
- d. What is the relationship between the area of the rectangle and the obtuse triangle?
Answers will vary. The area of the rectangle is double the area of the obtuse triangle. The area of the obtuse triangle is half the rectangle.
- e. Based on this relationship, what is the area of the obtuse triangle?
 6 cm^2



Use inquiry-based learning to help students draw connections between parallelograms, rectangles, and triangles. This will build conceptual understanding as students reason about how to find the area of a triangle.

- What do you know about parallelograms?
- What is the area of the parallelogram?
- How do the diagrams show relationship between the area of a parallelogram and the area of a rectangle?
- How is the height of a parallelogram related to the height of an obtuse triangle?



Consider using manipulatives, such as tangrams, to rearrange the parallelogram into a rectangle. This will help students who have trouble with spatial reasoning. What do you notice about the height and base for the rectangle and triangle? What do you wonder about their areas?

7.1.4 Bring It Together: Formula for the Area of a Triangle

Component Overview: Students synthesize the information they learned in the 7.1.2 Exploration and 7.1.3 Collaboration to create concise definitions and formulas for the area of a right, acute, and obtuse triangle.



7.1.4 Bring It Together: Formula for the Area of a Triangle

1. Complete the statements.

The area of the **right** triangle was ☐ half ☐ the same as the area of the rectangle with the same base and height.

The area of the **acute** triangle was ☐ half ☐ the same as the area of the rectangle with the same base and height.

The area of the **obtuse** triangle was ☐ half ☐ the same as the area of the rectangle with the same base and height.

2. The formula for the area of any rectangle is $A = bh$.

$$A = \text{base} \cdot \text{height}$$

The formula for the area of any triangle is $A = \frac{1}{2}bh$.

$$A = \frac{1}{2}(\text{base} \cdot \text{height})$$



Use this section to encourage synthesis, not memorization. Consider using this section as notetaking for students during the 7.1.3 Collaboration activity. Students may need to draw a small diagram next to each description to help visualize their thinking.



Alternatively, if time permits, prompt students to write a note to an absent student, a sibling, or a family member explaining how they can use rectangles to find the area of triangles.

7.1.5 Wrap-Up: Reflection

Component Overview: In this Wrap-Up, students will reflect on their understanding of the lesson. Students will place an X on the line to visually describe their understanding of the lesson. Students will explain their placement to self-reflect on what they know and what they need to improve on.



7.1.5 Wrap-Up: Reflection

1. Place an **X** on the line to indicate your mastery of today's lesson.

I can explain why the area of a triangle is half the area of a rectangle with the same base and height for right triangles, acute triangles, and obtuse triangles.

I need help.	I'm getting	I understand and
I can't get started.	there.	have evidence.



2. Write a brief explanation of your placement of the X.

Answers will vary.

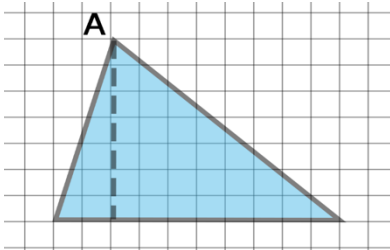


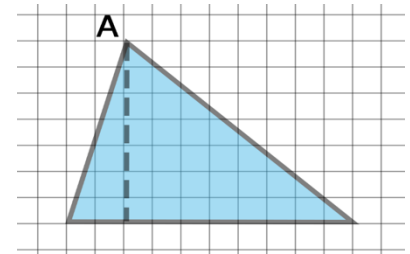
For students who might struggle representing their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example.

7.2 Area of Triangles – Part 2

Lesson Introduction

 Benchmark	MA.6.GR.2.1 Derive a formula for the area of a right triangle using a rectangle. Apply a formula to find the area of a triangle.		 Learning Target	I can apply a formula to find the area of triangles.
 Benchmark Coherence	Building On MA.5.GR.2.1 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas.		Working Toward MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi.	
 Essential Questions	- Where is the base of a triangle in relation to the height? - How can you use the base and height of a triangle to find the area? - How can you find the height of an acute or obtuse triangle?			
 Lesson Narrative	In this lesson, students use the formula for the area of a triangle to solve for the area of triangles. The lesson starts by making sure students can correctly identify the base and height of triangles. Once students have an understanding on how to find the base and height, they substitute values into the formula to solve for the area of a triangle.			
 Component Summary	7.2.1 Warm-Up (~5 min.)	Find the area of a triangle and rectangle and describe their relationship (<i>SE p. 303</i>)	7.2.3 Guided Practice (10 min.)	Identify base and height from given measurements on a triangle and calculate the area (<i>SE p. 305</i>)
	7.2.2 Guided Instruction (20-25 min.)	Find base and height by finding perpendicular lines in the given triangle (<i>SE p. 303</i>)	7.2.4 Your Turn! (10 min.)	Identify base and height from given measurements on a triangle and calculate the area (<i>SE p. 306</i>)
	7.2.5 Wrap-Up (~5 min.)	Identify base and height from an obtuse triangle to calculate area (<i>SE p. 307</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Base - Height - Area of a triangle <u>Additional Terms:</u> - Perpendicular		(Optional) Straight edge (Optional) Graph paper - Four-Function Calculator	N/A

 Teacher Tips	Common Misconceptions - Students may identify lines that are perpendicular to the height rather than the base (like answer choice c in 7.2.3 Guided Practice). - Students may incorrectly identify a non-perpendicular side of a triangle as the base or height. - Students may get stuck while substituting values into the area of a triangle formula.	Things to Consider - Take time to discuss how base is not always the bottom and height is not always the vertical side. - When given more than two measurements on a triangle, encourage students to label the height and base for each problem. - When needing to draw the height outside of an obtuse triangle, remember that the base does extend to the drawn line. - Consider allowing students to have access to four-function calculators.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Clarify, Critique, Correct Consider utilizing a clarify, critique, correct learning routine for 7.2.3 Guided Practice. Provide students with the following statement: The base of a triangle is the bottom line, and the height is the longest line. Prompt students to discuss the problems with the given statement and how they could rewrite.	MA.K12.MTR.6.1: Reasonableness This lesson provides multiple opportunities for students to check their calculations and determine the reasonableness of their solutions. Encourage students to review their work as they use the formula to find the areas of the triangles. This will reinforce the importance of checking their work as they progress through tasks and encourage mathematical precision.
 Differentiate Instruction	Utilize graphing paper or similar resources as scaffolding for students to recognize the perpendicular relationship between the base and the height. By drawing the triangle on the grid the student can use the gridlines on a coordinate plane to draw the height Consider modeling this approach in 7.2.2 Guided Instruction and prompting students to use this method during 7.2.3 Guided Practice.	
		
Lesson Notes:		

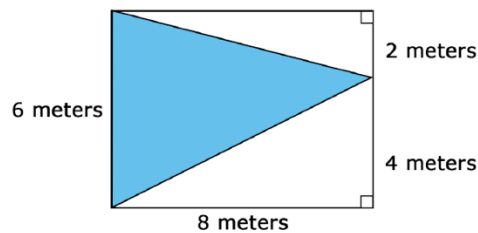


7.2.1 Warm-Up: Bell Work

Component Overview: In this Warm-Up, students find the areas of a triangle and a rectangle using dimensions from a diagram of a triangle inscribed in a rectangle.



7.2.1 Warm-Up: Bell Work



1. What is the area of the rectangle? Include units.
48 m²
2. What is the area of the shaded triangle? Include units.
24 m²
3. Explain the relationship between the area of a rectangle and triangle that have the same height and base.
The area of a triangle is half the area of a rectangle with the same height and base.



Use the bell work to inform instruction and student misconceptions. Consider reviewing student work to notice if students used the formula for question 2 or if they found $\frac{1}{2}$ of the area of the rectangle. If students struggled, consider using small group instruction to help students with their misconceptions.

7.2.2 Guided Instruction: Find the Area of a Triangle

Component Overview: In this Guided Instruction, students explain how base and height are related within a triangle. Students use the given base and height to calculate the area of a triangle.



7.2.2 Guided Instruction: Find the Area of a Triangle

To find the area of any triangle, identify a base of the triangle and its corresponding height.

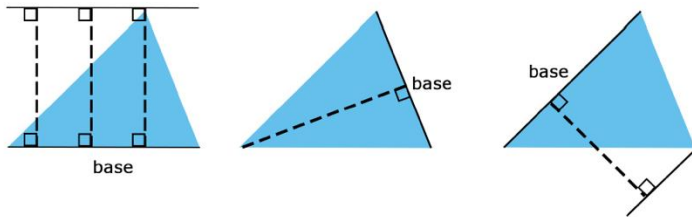


Any of the three sides of a triangle can be the **base**. The term “base” can refer to the side of a triangle and also its measurement.



The corresponding **height** of a triangle is the length of the perpendicular segment drawn from a base to its opposite vertex. The opposite vertex is the vertex of the triangle that is *not* an endpoint of the base.

Examples of bases and their corresponding heights on copies of the same triangle are shown. Each dashed line segment represents the height of the triangle.



1. In each example the height is _____ to the base.

- ☐ perpendicular
- ☐ opposite

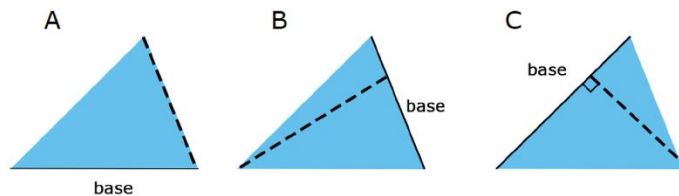


This Guided Instruction is intentionally scaffolded to solidify student understanding before applying the formula to find the area of a triangle. Consider having a student read each blue box and go through each example provided. Encourage students to ask questions as the class reviews each example. Guide the discussion with questions such as:

- What does “perpendicular” mean?
- What is the difference between base and height?
- Why do some of the heights in the triangles go outside out of the triangles?

7.2.2 Guided Instruction: Find the Area of a Triangle (continued)

2. The side used for the base is labeled on three triangles below. Complete the table to explain why each of the dashed lines do not represent the height of each triangle.



Triangle	Explanation
Triangle A	<i>The base is not perpendicular to the height.</i>
Triangle B	<i>The base does not form a 90° angle with the height.</i>
Triangle C	<i>The height does not reach the highest point of the triangle.</i>



Area of a Triangle $= \frac{1}{2}bh$,
where b is the base and h is the height.



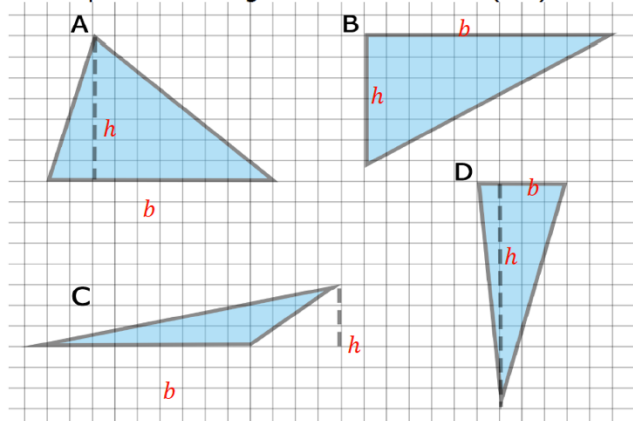
Question 2 intentionally shows bases using different orientations. Make it a practice when working with students to vary the position of the base. Any side of a triangle can be called the base if there is a perpendicular that is drawn from the base to its opposite vertex to represent the height.



Consider having students who struggle visually to draw and cut out separate triangles with lines drawn like each option so that they can rotate each one to have a better visual when the base is on the bottom.

7.2.2 Guided Instruction: Find the Area of a Triangle (continued)

3. Four triangles are shown on the grid. The side length of each square on the grid is 1 centimeter (cm).



- For each triangle, label the line segment that is drawn to represent the height, h . Then, label the side that is the base, b . *Answer shown on grid.*
- In the table, record the measurement for the base and corresponding height for each triangle. Then, find the area of each triangle. Each square on the grid represents 1 centimeter (cm) by 1 cm.

Triangle	Base (cm)	Height (cm)	Area (cm ²)
Any Triangle	b	h	$\frac{1}{2}bh$
Triangle A	10 cm	7 cm	35 cm ²
Triangle B	11 cm	6 cm	33 cm ²
Triangle C	10 cm	3 cm	15 cm ²
Triangle D	4 cm	11 cm	22 cm ²



For question 3b consider completing the table for Triangle A then releasing students to complete the table while circulating listening for student misconceptions.

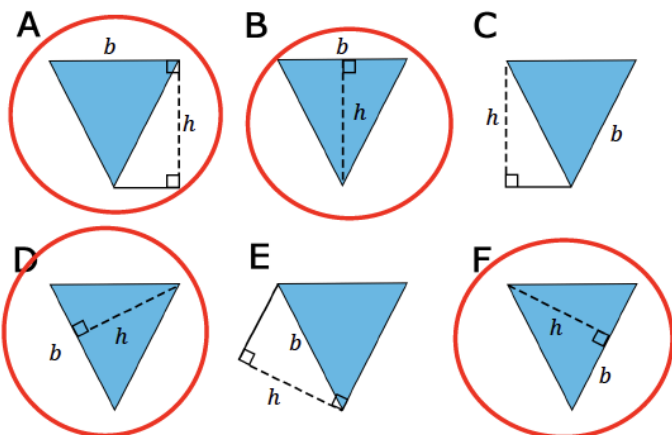
7.2.3 Guided Practice: Find the Area of a Triangle

Component Overview: In this Guided Practice, students work with a partner to identify correctly labeled base and height within a triangle and use that base and height to find the area of a triangle.



7.2.3 Guided Practice: Find the Area of a Triangle

- Discuss with your partner which drawings correctly label base, b , and its corresponding height, h , represented by the dashed line segment. Circle all the correct drawings. *Answer shown below.*



Clarify, Critique, Correct

Have students read the following sentence: “The base of a triangle is the bottom line, and the height is the longest line of a triangle.”

Have students discuss what the problems are with this statement and how they could rewrite. If multiple students are getting different answers, position them to turn and talk about why they think they are right with their partner and foster a discussion for students to arrive at the same answer.

7.2.3 Guided Practice: Find the Area of a Triangle (continued)

2. For each triangle below, circle the base measurement and put a square around the height measurement. Then, use the base and height to find the area of each triangle. Show your work. Include units.

	Area = <u>90 cm²</u>
	Area = <u>87.75 m²</u>
	Area = <u>425.25 mm²</u>
	Area = <u>30 cm²</u>



Consider having students first circle the base and square the height and then come back together as a class to check in before finding the area. Ask questions such as: What helped you determine the height? What helped you determine the base?



Lead students to understand that the marking of the right angle can be helpful in denoting which side is the base.



Consider having students write the units for the area on the answer line before they solve the problem to not forget or write the incorrect ones.

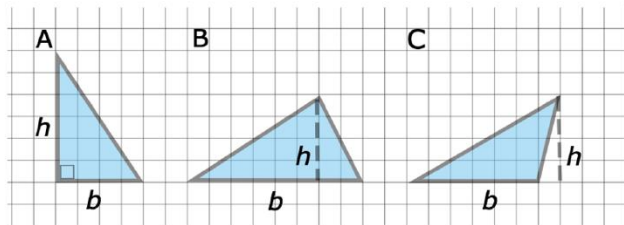
7.2.4 Your Turn!

Component Overview: In this Collaboration, students use the given height and base of each triangle to find the area of each triangle. Students decipher which measurements are used for base and height and which are not.



7.2.4 Your Turn!

- For each triangle, a base and its corresponding height are labeled. Complete the table by finding the area of each triangle. The side length of each square on the grid is 1 inch (in.).

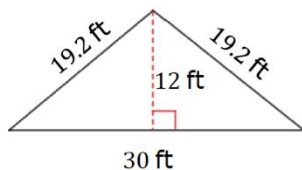


Triangle	Area (in ²)
A	12 in ²
B	16 in ²
C	12 in ²



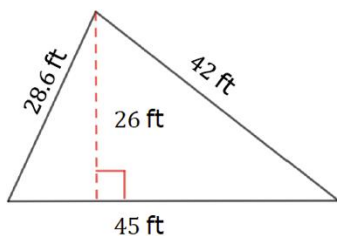
Relate measuring the height of an object to when you measure your own height. Relate what perpendicular lines look like by having students stand and describe their bodies' relationships to the floor. Having students use their arms to show perpendicular lines will engage kinesthetic learners.

- Find the area of each triangle. Include units.



Area = 180 ft²

b.



Area = 585 ft²

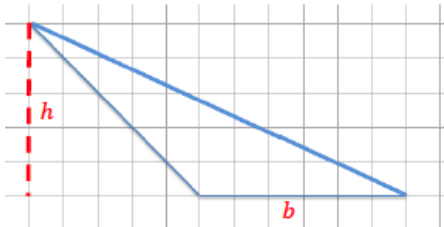
7.2.5 Wrap-Up: Ticket Out the Door

Component Overview: In this Wrap-Up, students find the base and height measurements of a triangle and calculate the area.



7.2.5 Wrap-Up: Ticket Out the Door

1. A diagram of a triangle is shown on a grid. The side length of each square on the grid represents 1 foot (ft).



- a. Label a base and its corresponding height for the triangle.

Shown on image.

- b. Find the area of the triangle. Include units.

15 ft²



Use the bell work and the ticket out the door to inform you of student progression on misconceptions. If students struggled, consider using small group instruction to help students with their misconceptions.

7.3 Decomposing Quadrilaterals to Find Area

Lesson Introduction

 Benchmark	MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.		 Learning Target	I can find the area of quadrilaterals by decomposing them into triangles and rectangles.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.GR.2.2 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas.		MA.7.GR.1.2 Solve mathematical or real-world problems involving the area of polygons or composite figures by decomposing them into triangles or quadrilaterals.	
 Essential Questions	- How can the decomposition of a figure help you find missing dimensions? - How can you use the area of triangles and rectangles to find the area of a quadrilateral? - Is there more than one way to decompose a quadrilateral?			
 Lesson Narrative	In this lesson, students use what they have learned previously in the unit to decompose quadrilaterals into familiar shapes to determine their area. Students begin by finding the area of a parallelogram by decomposing it into triangles. This strategy is then applied to different quadrilaterals as students decompose them into triangles to find the area. This understanding is extended as students explore the different ways in which shapes can be broken apart to find their area and find the sum of the areas of the smaller shapes to determine the area of the whole.			
 Component Summary	7.3.1 Warm-Up (~5 min.)	Find perimeter and area of a rectangle (<i>SE p. 308</i>)	7.3.4 Guided Practice (10 min.)	Decompose quadrilaterals into rectangles and triangles to find their area (<i>SE p. 311</i>)
	7.3.2 Collaboration (10 min.)	Find the area of triangles that make up parallelograms to find the area of a parallelogram (<i>SE p. 309</i>)	7.3.5 Your Turn! (10 min.)	More practice decomposing quadrilaterals in multiple ways into rectangles and triangles to find their area (<i>SE p. 312</i>)
	7.3.3 Exploration (15 min.)	Decompose quadrilaterals to find the area of shapes whose formulas are known (<i>SE p. 309</i>)	7.3.6 Wrap-Up (~5 min.)	Decomposing a quadrilateral to find the area (<i>SE p. 314</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Quadrilateral - Dimensions (length, width, height, base)		- Separate paper - Scissors - Glue - Four-Function calculator	- If you are doing the Collect and Display for 7.3.5 Your Turn! question 3, you may want to display student drawings under the document camera, on their desks, or around the room on the walls. - Consider precutting the triangles and rectangles to save time. - If differentiating by content, choose numbers to modify before beginning the lesson.

 Teacher Tips	Common Misconceptions - When decomposing shapes, students may not fully decompose a quadrilateral and get stuck with a shape they do not know how to find the area for. - Students may forget to solve for missing measurements they need to find the area and use another measurement that does not belong to the decomposed shape. - Students may incorrectly think that they must have a triangle and rectangle when decomposing.	Things to Consider - Students need to be reminded that their goal is to break down a quadrilateral into triangles and rectangles. - Encourage students to find more than one way to decompose the shapes. - Position students to notice that some quadrilaterals can only be broken down one way, while others will have multiple ways, including different types of shapes used and how many shapes can be used. - Monitor students and the time it takes them to determine how to decompose the quadrilaterals. If time is a problem, then assign the questions in Your Turn! for homework.
	Literacy and Language Routines Collect and Display Consider positioning students to describe a different way they could decompose the shape to a partner while the partner draws and labels the corresponding lengths. Select students to show their drawing to the class and explain how they decomposed their shape. Choose students who decomposed differently to share to give students a different perspective.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations Throughout this lesson, students represent multiple ways to dissect quadrilaterals into different shapes to find their areas. Students explore the ways in which shapes can be partitioned and put back together and connect this to the areas of familiar shapes, such as triangles and rectangles.
	 Differentiate Instruction	
Lesson Notes:		

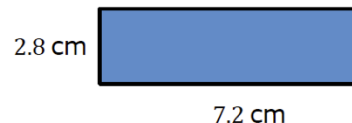
7.3.1 Warm-Up: Perimeter and Area of a Rectangle

Component Overview: Students are working independently to find the area and perimeter of a rectangle. This activity will help activate prior knowledge of perimeter and area of a rectangle, which will be built upon in this lesson.



7.3.1 Warm-Up: Perimeter and Area of a Rectangle

1. A rectangle is shown with side lengths given in centimeters (cm).



- a. Find the perimeter of the rectangle. Include units.

Perimeter = 20 cm

- b. Find the area of the rectangle. Include units.

Area = 20.16 cm²



Position students to write out the formulas for perimeter and area of a rectangle to help with substitution. Activate prior knowledge by prompting students to explain the difference between perimeter and area.

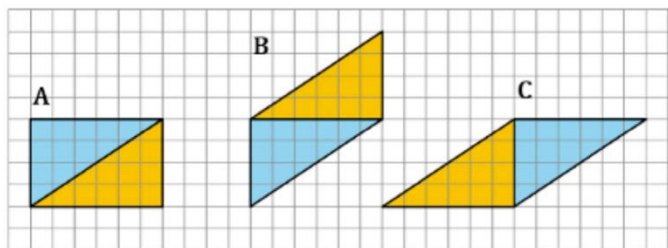
7.3.2 Collaboration: Decomposing Parallelograms

Component Overview: Students decompose parallelograms into triangles to find the area of each triangle. Students use the sum of the area of triangles to find the total area of the parallelogram. This positions students to discover how the area of more complex shapes can be found by being broken down into shapes that students already know how to find the area of.



7.3.2 Collaboration: Decomposing Parallelograms

Three parallelograms are shown on the grid. Each square is 1 centimeter (cm) by 1 cm.



1. Work with your partner to complete the table using the figures shown in the grid. Include units.

Parallelogram	Area of Blue Triangle	Area of Gold Triangle	Sum of the Two Triangles
A	12 cm^2	12 cm^2	24 cm^2
B	12 cm^2	12 cm^2	24 cm^2
C	12 cm^2	12 cm^2	24 cm^2

2. Discuss with your partner how the areas of the triangles are related to the area of the parallelogram they form. Write a brief summary of your discussion.

Answers will vary.

The triangle areas can be measured separately and added together to find the area of the parallelogram. If you move the triangles around, the parallelogram still has the same area.



As students work in pairs, circulate to monitor student progress and pose the following questions to deepen understanding and prime students for the rest of the lesson:

- If you rearranged the triangles, would the area still be the same?
- What do you notice about all the areas of the parallelograms?

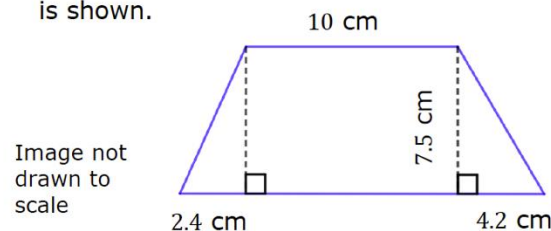
7.3.3 Exploration: Finding Area by Decomposing

Component Overview: Students will be introduced to finding the area of quadrilaterals by decomposing them into shapes in which they know how to find the area. Students will find the area of each shape and then find their sum, which will be the area of the quadrilateral.



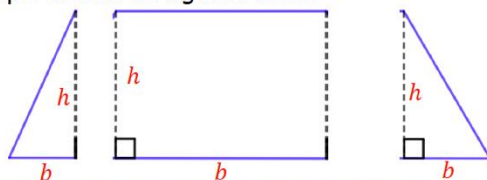
7.3.3 Exploration: Finding Area by Decomposing

1. A quadrilateral with measurements in centimeters (cm) is shown.



After students cut the quadrilateral but before they glue it down, consider giving students time to move the shapes together and apart a few times and letting them rearrange the shapes to see how they could make a quadrilateral in a different arrangement. Prompt students to share what they know about quadrilaterals.

- a. Copy the image on a piece of paper.
- b. Decompose the quadrilateral into two triangles and a rectangle by cutting along the dotted lines. Glue or tape the three figures here.



- c. Using the measurements given in the original image, label the base and height of each figure.
- d. What is the area of each figure? Include units.

Triangle 1: 9 cm^2

Rectangle: 75 cm^2

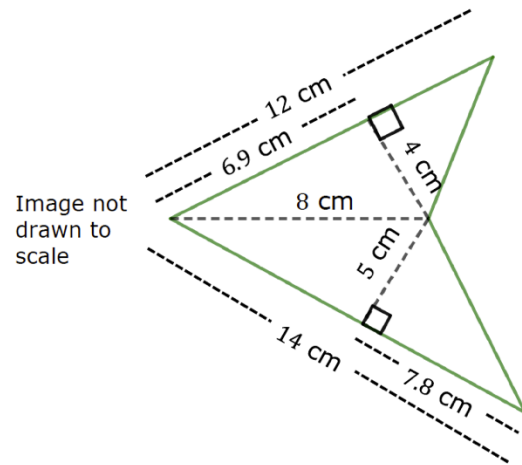
Triangle 2: 15.75 cm^2

- e. Find the area of the original quadrilateral by finding the sum of the three figures. Include units.

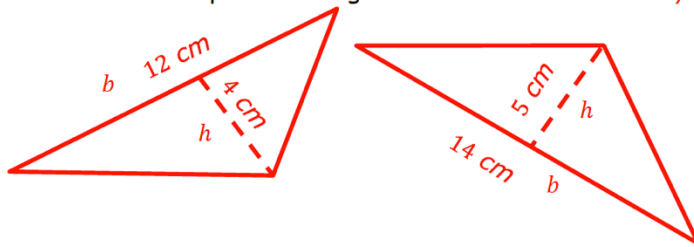
$$T1 + \text{Rect} + T2 = 9\text{cm}^2 + 75\text{cm}^2 + 15.75\text{cm}^2 = 99.75\text{cm}^2$$

7.3.3 Exploration: Finding Area by Decomposing (continued)

2. Another quadrilateral is shown.



- a. Decompose the quadrilateral into triangles. Sketch the decomposed triangles here. *Answers will vary.*



- b. Using the measurements given in the original image, label the base and height of each triangle.
Answer shown on triangle sketches.
- c. Find the area of the original quadrilateral by finding the sum of the areas of the decomposed triangles. Include units.

$$A_1 = \frac{12 \text{ cm} \times 4 \text{ cm}}{2} = 24 \text{ cm}^2 \quad A_2 = \frac{14 \text{ cm} \times 5 \text{ cm}}{2} = 35 \text{ cm}^2$$

$$A_T = A_1 + A_2 = 24 \text{ cm}^2 + 35 \text{ cm}^2 = 59 \text{ cm}^2$$



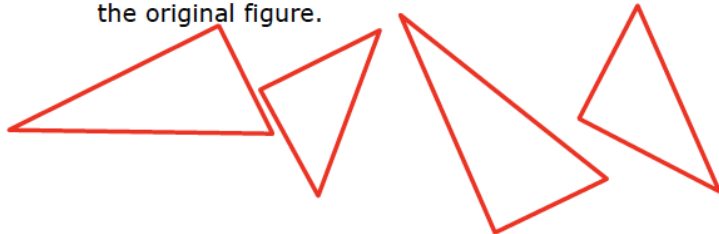
On a separate sheet of paper, have students draw three other ways they can decompose the quadrilateral in question 2. Have students share their answers with a partner to compare.



Overlapping figures are difficult for students to break apart. Ask students to share how they would break apart the quadrilateral. Different pairs may break the quadrilateral up differently. Encourage students to label the length of the sides in question 2a.

7.3.3 Exploration: Finding Area by Decomposing (continued)

- d. Sketch any other ways you could have decomposed the original figure.



- e. Explain if decomposing the figure changes the area of the original figure.

Answers will vary. Decomposing a figure will not change the original area because you are just breaking apart a figure. You are not expanding or shrinking it.

7.3.4 Guided Practice: Finding Area by Decomposing

Component Overview: Students will continue to practice decomposing quadrilaterals to find their areas. Students will also need to use deductive reasoning and subtraction to find missing side lengths to find the area of the shapes that make up the quadrilaterals.



7.3.4 Guided Practice: Finding Area by Decomposing

- Decompose the quadrilateral to find the area. Lengths are shown in feet (ft). Include units.

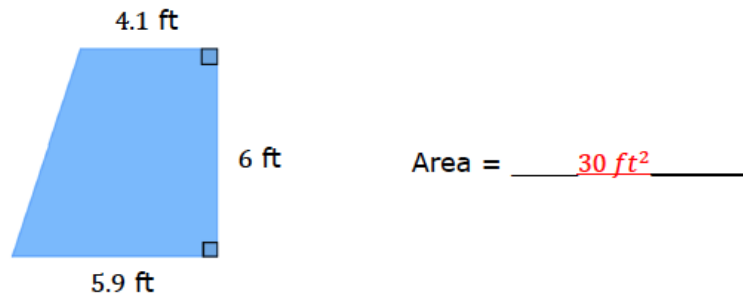
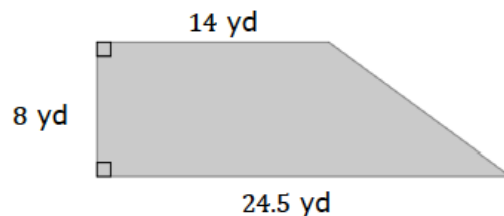


Image not drawn to scale

- The floor plan of a store is shown with measurements in yards (yd).



- The store found 160 square yards (yd²) of tile on sale. Based on the floor plan drawn, will the store have enough tiles to redo the flooring if they buy what is on sale?

Yes, the area of the floor is 154 yd².

- Determine the amount of tile, in yd², the store is short or has left over.

There are 6 yd² left over.



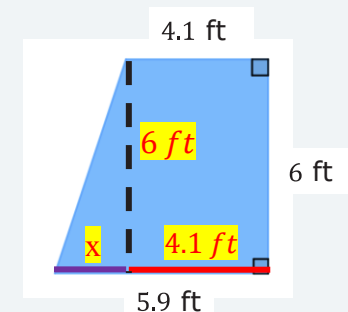
Notice and Wonder

Before students solve each problem, have them decompose the quadrilateral. Then have students participate in a Notice and Wonder routine.

- If you decompose the quadrilateral into a triangle and a rectangle, what do you notice about the measurements for base and height?
- How would you find the base of the triangle with the given base of the quadrilateral and the length of the rectangle?



Consider having students label the quadrilateral like the example below to visualize how to find the base of the triangle.



7.3.5 Your Turn!

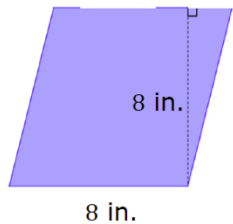
Component Overview: Students will continue to decompose quadrilaterals to find the area. In this activity, students will also need to calculate the missing base of a triangle within the shape using the given information to find the area. Encourage students to mark up and manipulate the quadrilaterals to get a better visual of the decompositions.



7.3.5 Your Turn!

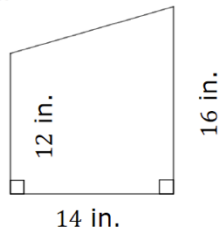
Find the area of each quadrilateral by decomposing. Lengths are shown in inches (in.). Include units.

1. 6 in. 2 in.



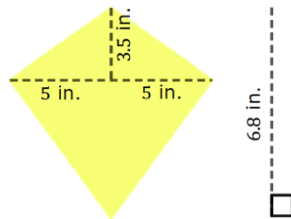
$$\text{Area} = \underline{64 \text{ in}^2}$$

2.



$$\text{Area} = \underline{196 \text{ in}^2}$$

3.



$$\text{Area} = \underline{34 \text{ in}^2}$$

Image not drawn to scale



Students may incorrectly think that because there is already a line drawn for the triangle on the right side of question 1 that the quadrilateral will only be decomposed into two shapes. Position students to discuss how this quadrilateral can be further decomposed.

- What are two ways you could figure out that a quadrilateral is not a square or a rectangle?
- Do you have enough information in question 1 to decompose the quadrilateral into two triangles and find their areas?
- Do you have enough information in question 2 to decompose the quadrilateral into two triangles and find their areas?



Consider having students use manipulatives to create and decompose these shapes. Students could also draw the shapes on paper or create the quadrilaterals using tangrams.



Collect and Display

Consider positioning students to describe a different way they could decompose the shape to a partner while the partner draws and labels the corresponding lengths. Select students to show their drawing to the class and explain how they decomposed their shape. Choose students who decomposed differently to share to give students a different perspective.

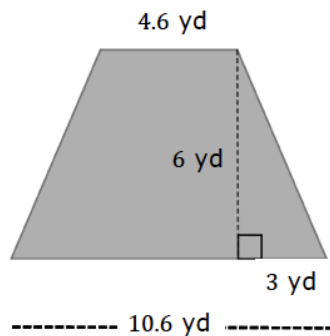
7.3.6 Wrap-Up: Ticket Out the Door

Component Overview: Students will demonstrate their understanding of decomposing quadrilaterals and finding their areas. Students will need to decompose the quadrilateral into two triangles and a rectangle.



7.3.6 Wrap-Up: Ticket Out the Door

- Find the area of the quadrilateral by decomposing. Lengths are given in yards (yd). Include units.



Area = 45.6 yd²

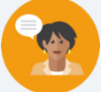


Consider requiring students to draw how they decomposed the quadrilateral and to label the decomposition. Their drawing will help determine common misconceptions students may be having and inform your instruction.

7.4 Deriving Area Formulas for Trapezoids, Parallelograms, and Rhombi

Lesson Introduction

 Benchmark	MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi.		 Learning Target	I can use the formulas for the area of triangles and rectangles to derive formulas for the area of trapezoids, parallelograms, and rhombi.
 Benchmark Coherence	Building On MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.		Working Toward MA.912.GR.4.4. Solve mathematical and real-world problems involving the area of two-dimensional figures.	
 Essential Questions	How do you derive the area formulas for parallelograms, trapezoids, and rhombi by decomposing and rearranging their pieces?			
 Lesson Narrative	This is part one of a two-part lesson series on the areas of parallelograms, trapezoids, and rhombi. In this lesson, students derive the formulas for the area of parallelograms, trapezoids, and rhombi by decomposing and rearranging their pieces. This work will prepare students for the next lesson, where they will be applying the formulas to solve real-world problems.			
 Component Summary	7.4.1 Warm-Up (~5 min.)	Notice and Wonder about a decomposed rectangle (<i>SE p. 315</i>)	7.4.4 Activity (10 min.)	Decompose trapezoids to find the formula for area of trapezoids (<i>SE p. 318</i>)
	7.4.2 Refresher (5 min.)	Review decomposing to find area of a triangle (<i>SE p. 315</i>)	7.4.5 Activity (10 min.)	Decompose rhombi to find the formula for area of rhombi (<i>SE p. 319</i>)
	7.4.3 Activity (10 min.)	Decompose parallelograms to find the formula for area of parallelograms (<i>SE p. 316</i>)	7.4.6 Wrap-Up (~5 min.)	Self-reflection on what was most difficult or confusing (<i>SE p. 321</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Quadrilateral - Parallelogram - Trapezoid - Rhombus <u>Additional Terms:</u> N/A		(Optional) Graph paper (Optional) Colored highlighters (Optional) Scissors	Prepare copies of directions and problems for the Activities (7.4.3-7.4.5) and set up stations for each activity. Determine groups for rotations through the different stations. Consider using data and/or observations from formative assessments to determine grouping arrangements.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse which diagonal represents the base and which diagonal represents the height in the triangles created when decomposing a rhombus. - Students may not recognize that the bases in a trapezoid can be used interchangeably in the formula.	- The activities provide scaffolding, yet students may still struggle with deriving the formulas. Allow for some productive struggle. Students may struggle with the rhombus activity the most, so consider any extra help your students may need to complete the activity. - Students without a strong understanding of equivalent algebraic expressions may think their “formula” does not match their peer’s formula or any provided formulas in the unit. When supporting these students, use correct mathematical properties to explain why their formula is the same. For example, if a student has $\frac{1}{2}d_2d_1$ instead of $\frac{1}{2}d_1d_2$, explain that the commutative property shows the two are equivalent. Free students to use their own derived formulas for the rest of the unit. A student may understand $A = \frac{1}{2}b_1h + \frac{1}{2}b_2h$ more than $A = \frac{1}{2}h(b_1 + b_2)$.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Notice and Wonder	MA.K12.MTR.2.1: Multiple Representations
	The Warm-Up and Refresher purposefully position students to confidently share their observations about a decomposed rectangle. The discussion will lay the foundation for their explorations with decomposing parallelograms, trapezoids, and rhombi to derive their area formulas.	In the Activities (7.4.3-7.4.5), students decompose parallelograms, trapezoids, and rhombi. They rearrange the decomposed pieces to provide a concrete representation of the area formula for the figure. Through the activities, students are guided from concrete, to pictorial, to abstract representations as their understanding of area progresses.
Lesson Notes:		

7.4.1 Warm-Up: Notice and Wonder

Component Overview: Students reflect on what they notice and wonder about an image of a rectangle decomposed into two triangles and a trapezoid. Student thinking and reasoning in this Warm-Up lays the foundation for their work in the subsequent activities.



7.4.1 Warm-Up: Notice and Wonder

Two figures are shown.

Figure 1

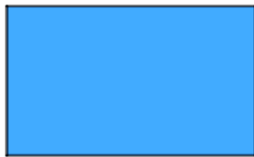


Figure 2



1. What do you notice about these two figures?
Answers will vary. I notice that when combined, Figure 2 creates Figure 1.
2. What do you wonder about the figures?
Answers will vary. I wonder what shape is in the middle of the two triangles in Figure 2.



Notice and Wonder

There are no wrong answers. This fosters an environment where each student feels safe to participate. After sharing, students will likely feel more confident sharing their observations during their explorations as they derive the area formulas in subsequent activities.



How can you combine the two triangles in Figure 2 to make a smaller rectangle? How could the shape in the middle of Figure 2 be decomposed into two smaller triangles?

7.4.2 Refresher: Decomposing to Find the Area of a Triangle

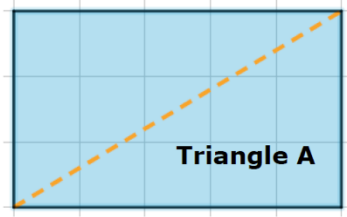
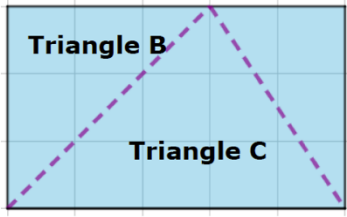
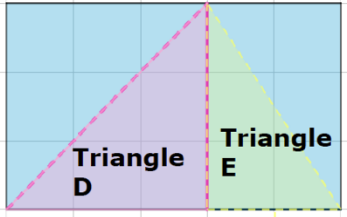
Component Overview: Students use three methods to decompose rectangles into triangles and then rearrange the triangles to find the area of the rectangle. Students can complete the table individually or with a partner. Debrief by engaging students with a whole-class discussion to ensure each and every student is ready for the subsequent activities.



7.4.2 Refresher: Decomposing to Find the Area of a Triangle

Figures can be decomposed into smaller pieces that are then rearranged and composed into other shapes. Decomposing is useful when finding the area of a figure.

1. Complete the table.

The rectangle is divided into 2 triangles. The area of Triangle A is $\frac{1}{2}$ of the area of the rectangle.	
The rectangle is divided into 3 triangles. The area of Triangle B is $\frac{3}{10}$ of the area of the rectangle. The area of Triangle C is $\frac{1}{2}$ of the area of the rectangle.	
The rectangle is divided into 4 triangles. The area of Triangle D is $\frac{3}{10}$ of the area of the rectangle. The area of Triangle E is $\frac{1}{5}$ of the area of the rectangle.	



Leverage this activity not as an opportunity to re-teach, but instead to highlight the utility of decomposition for finding area.



How do you know the area of triangle ____ is ____?
How could you check your answer?

7.4.3 Activity: Deriving the Formula for the Area of a Parallelogram

Component Overview: The next three activities can be used as three different stations. Groups of students can rotate between the stations in any order. For large classes, two copies of each station can be used. In this first activity, students decompose a parallelogram into a trapezoid and a triangle. Then, they shift the triangle to form a rectangle and derive the area formula for a parallelogram.



7.4.3 Activity: Deriving the Formula for the Area of a Parallelogram



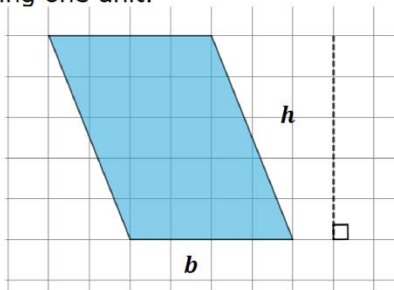
A **quadrilateral** is a polygon with exactly four sides and four vertices.

A parallelogram is a type of quadrilateral. Not only does a parallelogram have four sides and four vertices, but its opposite sides are also parallel.



A **parallelogram** is a quadrilateral containing two pairs of parallel sides.

1. A parallelogram is shown on a grid with each gridline representing one unit.



Complete each statement.

- a. The base of the parallelogram is 4 units.
- b. The height of the parallelogram is 5 units.



Consider your students and classrooms when planning the next three activities. Station rotations could be an interactive way for students to stay active while working.

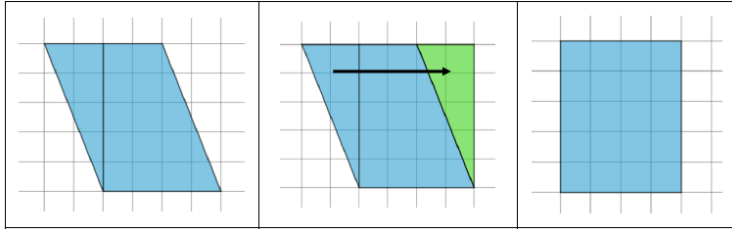
- Station 1: Activity 7.4.3 – Parallelogram
- Station 2: Activity 7.4.4 – Trapezoid
- Station 3: Activity 7.4.5 - Rhombus



Provide a kinesthetic entry point by having students physically draw, cut, and rearrange the parallelogram. Provide materials at each station or elect a “Materials Captain” per group to handle the supplies for each group.

7.4.3 Activity: Deriving the Formula for the Area of a Parallelogram (continued)

2. The images below show the same parallelogram being decomposed and then rearranged to form a rectangle.



Complete each statement.

- The base of the rectangle is 4 units.
 - The height of the rectangle is 5 units.
3. Discuss with your partner the relationship between the area of the parallelogram and the area of the rectangle. Write a summary of your discussion.
Answers will vary. The relationship between the area of the parallelogram and the rectangle is that the areas are the same.
4. Complete the statements.

A parallelogram can be decomposed and rearranged into a

☐ triangle
☒ rectangle
 with the same base and height. The

formula used to find the area of a rectangle is

- ☒ $A = bh.$
☐ $A = \frac{1}{2}bh.$

Therefore, the area of the parallelogram is

- ☒ $A = bh.$
☐ $A = \frac{1}{2}bh.$



Consider having students create a foldable and/or an Anchor Chart with images and examples of deriving each area formula to provide a visual entry point.

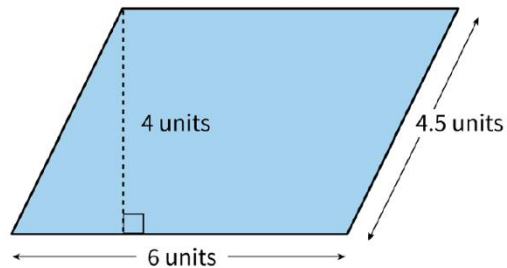


Consider asking students the following:

- How does the height of the parallelogram compare to the height of the rectangle?
- How does the base of the parallelogram compare to the base of the rectangle?
- How do these relationships help you to determine the area of the parallelogram?

7.4.3 Activity: Deriving the Formula for the Area of a Parallelogram (continued)

5. A parallelogram with its dimensions labeled is shown.



Students may use 4.5 as the height instead of 4 as the height of the parallelogram. Have students show how the triangle can be shifted to create a rectangle, then ask the student which value is the height of the rectangle.

- a. Fill in the blanks to find the area of the parallelogram.

$$\text{Area of parallelogram} = b \cdot h$$

$$\text{Area} = 6 \cdot 4$$

$$\text{Area} = 24 \text{ units}^2$$

- b. Three lengths are shown in the figure. With your partner, discuss the length that was not used to find the area of the parallelogram. Then, write an explanation for why this length was not used.
Answers will vary. The length on the slanted side is not perpendicular to the base therefore it is not the height, and the height must be used to find the area.

7.4.4 Activity: Deriving the Formula for the Area of a Trapezoid

Component Overview: Students decompose a trapezoid into two triangles and use the triangles to derive the area formula for trapezoids. Encourage students to discuss their strategies with a partner.



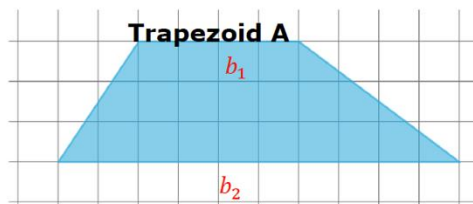
7.4.4 Activity: Deriving the Formula for the Area of a Trapezoid

A trapezoid is another type of quadrilateral.



A **trapezoid** is a quadrilateral with at least one pair of parallel sides.

Trapezoid A is shown on a grid with each gridline representing one unit.



The two parallel sides of a trapezoid are both called bases. To distinguish between the two bases, one is named base one, b_1 , and the other base two, b_2 .

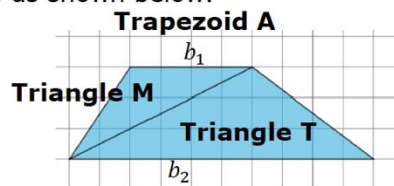
1. On Trapezoid A, label one of the bases b_1 and the other b_2 .
Shown on diagram.
2. Complete each statement.
 - a. The length of b_1 of Trapezoid A is 4 units.
 - b. The length of b_2 of Trapezoid A is 10 units.
 - c. The height of Trapezoid A is 3 units.



Clarify that students can label either base as b_1 and b_2 . Their labeling choices will impact their problem 2 answers.

7.4.4 Activity: Deriving the Formula for the Area of a Trapezoid (continued)

3. Trapezoid A is decomposed into two triangles and labeled, as shown below.



- a. Complete the statements.
The base of Triangle M, b_1 , is 4 units, and its corresponding height is 3 units.
The base of Triangle T, b_2 , is 10 units, and its corresponding height is 3 units. When a trapezoid is decomposed into two triangles by drawing a diagonal, the height of both triangles will always be the same.
- b. Davita wrote a formula for the area of Trapezoid A.

Davita's Equation

Area Trapezoid A = Area Triangle M + Area Triangle T

$$A = \frac{1}{2}b_1h + \frac{1}{2}b_2h$$

Explain how Davita determined this formula.

Answers will vary. Davita realized that the area of the trapezoid must equal the area of the two triangles; so she added together the formulas for the areas of both of the triangles to get the area of the trapezoid.

- c. The formula Davita wrote can be used to find the area of any trapezoid. Complete the steps described in the table to rewrite the formula.

Area of Trapezoid A	
Davita's formula	$A = \frac{1}{2}b_1h + \frac{1}{2}b_2h$
Use the distributive property to rewrite with a common factor h	$A = h\left(\frac{1}{2}b_1 + \frac{1}{2}b_2\right)$
Use the distributive property to rewrite with a common factor $\frac{1}{2}$	$A = \frac{1}{2}h(b_1 + b_2)$



Prompt students to make connections to their decomposing in 7.4.2 Refresher.

- How is this the same as the Refresher?
- How is it different?
- What does it make you think about what the Area formula might be?



Students may struggle with understanding that the distributive property is applied. This is because the distributive property is often taught as distributing a value not factoring a value. It may be helpful to show the students $10 + 15 = 5(2 + 3)$ as the distributive property. Properties should be taught and used "both ways". The algebra in rewriting Davita's formula should not be the focus of this station. If you think it will be a stumbling block for your students then provide them with the answers. It is only here to show students how what they derived is equivalent to the formula that is often provided.

7.4.5 Activity: Deriving the Formula for the Area of a Rhombus

Component Overview: Students derive the formula for the area of a rhombus by decomposing it into two triangles. Continue to encourage students to collaborate with partners, making sense of their discoveries as well as any connections to area.



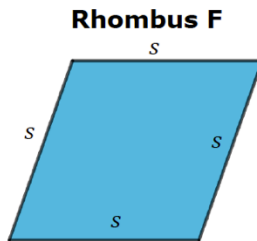
7.4.5 Activity: Deriving the Formula for the Area of a Rhombus

A rhombus is another type of quadrilateral.

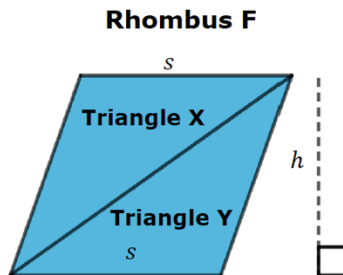


A **rhombus** is a quadrilateral containing four sides of equal length.

Rhombus F is shown with four congruent sides.



Rhombus F is decomposed into two triangles, Triangle X and Triangle Y, as shown.



Students may struggle with deriving the formula for the area of a rhombus using the two diagonals. The derivation of this formula extends student understanding both geometrically and algebraically. Be prepared with support for students as they complete this activity.



Prompt students to make connections to their decomposing in 7.4.2 Refresher.

- How is this the same as the Refresher?
- How is it different?

7.4.5 Activity: Deriving the Formula for the Area of a Rhombus (continued)

1. Felicia wrote expressions to represent the areas of the two triangles.

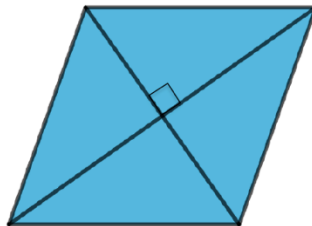
Area of Triangle X	Area of Triangle Y
$\frac{1}{2} \cdot s \cdot h$	$\frac{1}{2} \cdot s \cdot h$

Discuss with your partner whether you agree with the expressions Felicia wrote. Write a summary of your discussion.

Answers will vary. I agree with the expressions Felicia wrote because all of the sides are the same length on both triangles; which makes the triangles congruent and the formula to find area the same.

The diagonals of a rhombus can also be used to find its area.

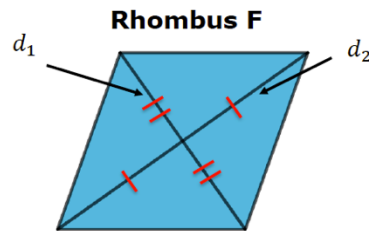
The two diagonals of a rhombus are perpendicular to each other and divide each other in half.



Is s the same for both triangles? How do you know? Is h the same for both triangles? How do you know? Could the formula for the rhombus be derived the same way the formula for parallelograms was derived in the 7.4.3 Activity?

7.4.5 Activity: Deriving the Formula for the Area of a Rhombus (continued)

2. The two diagonals, d_1 and d_2 , of Rhombus F are drawn and labeled below.



Mark on the figure the two lengths that are equal to $\frac{1}{2}d_1$ and the two lengths that are equal to $\frac{1}{2}d_2$.

3. The first column of the table shows Rhombus F with parts highlighted. Use the lengths marked on Rhombus F in the previous question to fill in the blanks to find the area of the triangles.

Triangle	Area
Rhombus F 	$\text{Area} = \frac{1}{2}(b)(h)$ $\text{Area} = \frac{1}{2}(d_2)(\frac{1}{2}d_1)$ Rewrite the area by multiplying any fractions. $\text{Area} = \frac{1}{4}d_2d_1$
Rhombus F 	$\text{Area} = \frac{1}{2}(b)(h)$ $\text{Area} = \frac{1}{2}(d_2)(\frac{1}{2}d_1)$ Rewrite the area by multiplying any fractions. $\text{Area} = \frac{1}{4}d_2d_1$

4. Compare your work with your partner and resolve any differences, if necessary.

Answers will vary.



Which diagonal represents the base of the triangles in the table? Which diagonal represents the height of the triangles in the table?



Consider having students use different colored highlighters to trace over d_1 and d_2 in the figures and ask them to label d_1 and d_2 in the table.

7.4.5 Activity: Deriving the Formula for the Area of a Rhombus (continued)

5. Work with your partner to express the area of Rhombus F as the sum of the area of the yellow triangle and the pink triangle found in the table.

$$\text{Area of Rhombus F} = \frac{1}{4}d_2d_1 + \frac{1}{4}d_2d_1 = \frac{1}{2}d_2d_1$$

6. Use words, values, or expressions from the bank shown to complete the statements. Items from the bank may be used more than once.

$\frac{1}{2}$

sum

$\frac{1}{4}(d_1)(d_2)$

diagonal

Drawing a **diagonal** of a rhombus divides the rhombus into two identical triangles. The area of the rhombus is equal to the **sum** of the areas of these two triangles. If one of these triangles is equal to $\frac{1}{4}(d_1)(d_2)$, then two of these triangles equal

$2 \cdot \frac{1}{4}(d_1)(d_2)$. Therefore, the formula for the area of a rhombus using its diagonals can be written as

$$A = \frac{1}{2}(d_1)(d_2).$$



Encourage students to write out their work in problem 5. Prompt students to look for like terms when adding. Remind students to look back at problem 3 when completing problem 6. Have students pay particular attention to the area column, as it highlights the area of one of the triangles created by drawing in a diagonal.



Consider asking students who finish early to apply the formulas they developed to create problems involving the areas of parallelograms, trapezoids, or rhombi. Then, have students swap problems with a partner and solve their partner's problems.

7.4.6 Wrap-Up: Reflection

Component Overview: Students reflect on the most difficult or confusing point in the lesson. Use data from this formative assessment to determine which students need remediation before the next lesson.



7.4.6 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point for you in today's lesson on deriving the formula of parallelograms, trapezoids, and rhombi?



Answers will vary. The muddiest point for me in today's lesson is deriving the formula for rhombi.



Consider allowing students to use an audio recorder (such as Flip Grid) or speech to text software if they struggle expressing themselves in writing. Students could also create a code (such as 0 = do not understand, 1 = sort of understand, 2 = completely understand) and use it to label the problems in the lesson.

7.5 Finding the Area of Trapezoids, Parallelograms, and Rhombi

Lesson Introduction

 Benchmark	MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi.		 Learning Target	I can find the area of trapezoids, parallelograms, and rhombi.
 Benchmark Coherence	Building On MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.		Working Toward MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.	
 Essential Question	How do you find the areas of trapezoids, parallelograms, and rhombi in mathematical and real-world problems?			
 Lesson Narrative	Students find the areas of trapezoids, parallelograms, and rhombi using the formulas they developed in the previous lesson. The lesson prepares students for the next three lessons, where they will find the areas of composite figures.			
 Component Summary	7.5.1 Warm-Up (~10 min.)	Watch and reflect on video on theme park engineer career (<i>SE p. 322</i>)	7.5.3 Your Turn! (20 min.)	Solve real-world area problems (<i>SE p. 324</i>)
	7.5.2 Guided Practice (15 min.)	Recall and apply area formulas (<i>SE p. 323</i>)	7.5.4 Wrap-Up (~5 min.)	Identify, correct errors, and find area of rhombi (<i>SE p. 325</i>)
 Resources	Glossary	Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Parallelogram - Trapezoid - Rhombus	(Optional) Graph paper (Optional) Pattern blocks (Optional) Chart paper & markers		- Warm-Up 7.5.1 contains a video to accompany the question prompts. - Consider using data and/or observations from the previous and current lessons to determine small group intervention that can occur during 7.5.3 Your Turn!

 Teacher Tips	Common Misconceptions - Students may use the diagonal length instead of the perpendicular length for the height. - Students may neglect to use the order of operations when calculating the areas of trapezoids using the formula $A = \frac{1}{2}h(b_1 + b_2)$. - Students may not include the units with their answers or may not raise the units to the power of 2.	Things to Consider - Create and display an Anchor Chart with formulas, examples, and images. Show students how understanding the formulas prevents the need to memorize them. Include the various forms of the formulas that students derived in the previous lesson. - Consider allowing students to use calculators so they can focus on applying the formulas. - Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with rational numbers. For example, question 2 in 7.5.2 could be a no calculator problem but the questions in 7.5.3 could be calculator problems.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect 7.5.2 Guided Practice provides an opportunity for students to think about and discuss the similarities and differences in the area formulas. Students also discuss how the formulas connect to the physical characteristics of the shapes.	MA.K12.MTR.7.1: Real-World Contexts Students apply the area formulas for trapezoids, parallelograms, and rhombi to solve real-world problems in 7.5.3 Your Turn! Student interest and motivation will be ignited as they see the utility of the area formulas.
 Differentiate Instruction	Consider providing students with graph paper or pattern blocks so they can physically model the real-world parallelograms, trapezoids, and rhombi in 7.5.3 Your Turn! Alternatively, challenge students to a “Rhombus Scavenger Hunt” in which they must find and document as many different rhombi as possible as homework or an extension project if including dimensions and area.	
Lesson Notes:		

7.5.1 Warm-Up: A Day in the Life

Component Overview: Students begin by watching the career profile video on theme park engineers. After watching the 3:11 minute video, prompt students to independently reflect on the prompt. Consider having a few students share their responses with the whole class.



7.5.1 Warm-Up: A Day in the Life

Betsy McIver, a mechanical engineer, works in the Walt Disney Imagineering area and designs systems for the interactive attractions at the theme park. She mentions that her job can include creative left-brain thinking, as well as technical right-brain thinking.

She says that to be successful at her job, it is important to understand math, design, and theatre. She also mentions that many disciplines, such as designers and architects, all work together to make everything possible in the entertainment engineering industry.



Consider inviting a guest speaker, such as the technology teacher, an engineer, designer, or architect to join the class who can answer students' questions about their careers and the mathematics they use.

1. Describe a ride you saw that looked exciting and entertaining.

Answers will vary. I saw a ride that went through Monsters Inc. that looked exciting.

7.5.2 Guided Practice: Area of Parallelograms, Trapezoids, and Rhombi

Component Overview: Students practice calculating the areas of parallelograms, trapezoids, and rhombi. Consider using formative assessment data from the previous lesson to determine which students will be most successful working independently, with a pair, or in a small group that you assist. Monitor which students are using different strategies so you can ask them to share during the whole-class debrief.



7.5.2 Guided Practice: Area of Parallelograms, Trapezoids, and Rhombi

- Complete the table to match each type of quadrilateral with the formula to find its area.

$$A = \frac{1}{2}h(b_1 + b_2) \quad A = \frac{1}{2}d_1d_2 \quad A = bh$$

Quadrilateral	Formula for the Area
parallelogram	$A = bh$
trapezoid	$A = \frac{1}{2}h(b_1 + b_2)$
rhombus	$A = \frac{1}{2}d_1d_2$

- The dimensions of three quadrilaterals are given in the table. Label each figure shown with the given dimensions in centimeters (cm). Then, calculate the area of each figure using the appropriate formula. Show your work.

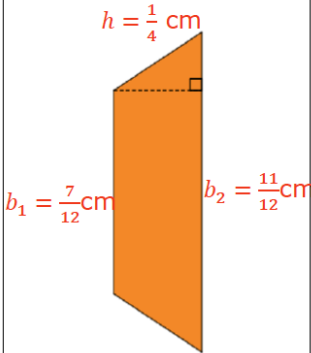
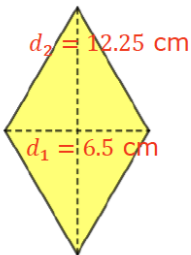
Figure	Dimensions	Area
Parallelogram $h = 8 \text{ cm}$ $b = 22 \text{ cm}$	$b = 22 \text{ cm}$ $h = 8 \text{ cm}$	$A = bh$ $A = 22 \cdot 8$ $A = 176 \text{ cm}^2$



Compare and Connect

Provide students with an opportunity to think about and discuss the similarities and differences in the area formulas. Students can also discuss how the formulas connect to the physical characteristics of the shapes. Allow students to add equivalent formulas from the previous lesson that they prefer to use.

7.5.2 Guided Practice: Area of Parallelograms, Trapezoids, and Rhombi (continued)

Figure	Dimensions	Area
Trapezoid  $h = \frac{1}{4} \text{ cm}$ $b_1 = \frac{7}{12} \text{ cm}$ $b_2 = \frac{11}{12} \text{ cm}$	$b_1 = \frac{7}{12} \text{ cm}$ $b_2 = \frac{11}{12} \text{ cm}$ $h = \frac{1}{4} \text{ cm}$	$A = \frac{1}{2} h(b_1 + b_2)$ $A = \frac{1}{2} \cdot \frac{1}{4} \cdot \left(\frac{7}{12} + \frac{11}{12} \right)$ $A = \frac{3}{16} \text{ cm}^2$
Rhombus  $d_2 = 12.25 \text{ cm}$ $d_1 = 6.5 \text{ cm}$	$d_1 = 6.5 \text{ cm}$ $d_2 = 12.25 \text{ cm}$	$A = \frac{1}{2} d_1 d_2$ $A = \frac{1}{2} \cdot 6.5 \cdot 12.25$ $A = 39.8125 \text{ cm}^2$



The orientation of figures may be difficult for some students. Help students experience figures drawn with different orientations by purposely drawing figures differently.

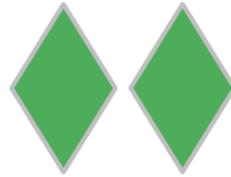
7.5.3 Your Turn!

Component Overview: Students practice finding the areas of parallelograms, trapezoids, and rhombi in a variety of real-world contexts. Students may engage with this activity independently or with a partner. Consider modifying the people, places, or things in the problems to contexts that appeal to your students' interests and everyday lives.



7.5.3 Your Turn!

- Alina is making rhombus-shaped earrings with green stones. The measure of one diagonal of each earring is $1\frac{3}{8}$ centimeters (cm), and the length of the other diagonal measures $1\frac{4}{9}$ cm. What is the area, in square centimeters (cm^2) of two earrings?



$$1\frac{71}{72}\text{ cm}^2$$

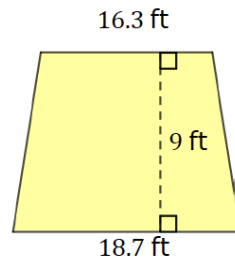


Consider pulling a small group during this activity to provide additional support to students based on observations from the previous and current lessons.



Does it matter which diagonal length you identify as d_1 and which you identify as d_2 ?

- One wall of Kel's apartment is shaped like a trapezoid with one base 18.7 feet (ft), the other base 16.3 ft, and a height of 9 ft.



Kel plans to paint this wall blue and needs to know how much paint to buy. What is the area of the wall Kel plans to paint?

$$157.5\text{ ft}^2$$

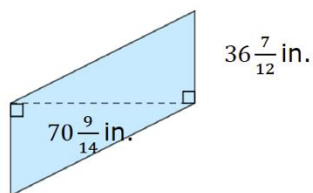


Why do you use square units when writing the area of rhombi, parallelograms, and trapezoids?

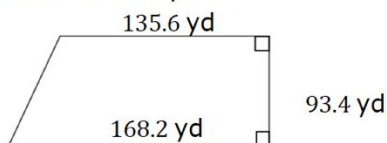
7.5.3 Your Turn! (continued)

3. Mika, a designer, is creating a geometric design in a tile floor. The design will be in the shape of a parallelogram with $36\frac{7}{12}$ inch (in.) bases and a height of $70\frac{9}{14}$ in. How many square inches of tile will she need for the design?

$$2,584\frac{59}{168}\text{ in}^2$$



4. The Smart family's horse pasture is in the shape of a trapezoid. One of the parallel sides is 168.2 yards (yd) long and the other is 135.6 yd long. The distance between these sides is 93.4 yd.



What is the area of the horse pasture?

$$14,187.46\text{ yd}^2$$

5. The Dockland Building in Hamburg, Germany is shaped like a parallelogram.

If the base of the building is 86 meters (m) and its height is 47 m, what is the area of the face of the building shown in the image?

$$4,042\text{ m}^2$$



In problems 3 and 5, does it matter which length you identify as the base and which you identify as the height?



Does it matter which base length you identify as b_1 and which you identify as b_2 ? Justify your response using numbers.



What does the question mean by "face" of the building? What shape is that?

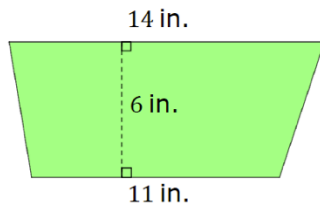
7.5.4 Wrap-Up: Ticket Out the Door

Component Overview: Students write a letter describing the correct and erroneous components of a sample of student work. Use data from this formative assessment to determine which students need remediation.



7.5.4 Wrap-Up: Ticket Out the Door

1. Don's work for finding the area of the trapezoid is shown below.



Don's Work
$A = \frac{1}{2} h(b_1 + b_2)$
$A = \frac{1}{2} (14)(11 + 6)$
$A = \frac{1}{2} (14)(17)$
$A = \frac{1}{2} \cdot 238$
$A = 119 \text{ in}^2$

- a. Write Don a note explaining what he did correctly and where he made an error.

Answers will vary. Don used the correct formula but he substituted the incorrect values for the base and the height and found the incorrect area.

- b. What is the area of the trapezoid?
75 in²



Students who struggle to express themselves in writing can circle the errors and then correct them or use an audio recorder (i.e. Flip Grid) or speech-to-text software.

7.6 Area of Composite Figures

Lesson Introduction

Lesson Introduction				
 Benchmark	MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.		 Learning Targets	- I can find the area of composite figures. - I can find the value of a missing dimension for a quadrilateral or a composite figure.
 Benchmark Coherence	Building On MA.5.GR.2.1 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas.		Working Toward MA.7.GR.1.2 Solve mathematical or real-world problems involving the area of polygons or composite figures by decomposing them into triangles or quadrilaterals.	
 Essential Questions	- How do you find the area of a composite figure? - How can you use the areas of triangles and rectangles to find the area of a composite figure? - How do you use given dimensions to find the missing dimension of a composite figure?			
 Lesson Narrative	This lesson progresses from refreshing students' prior knowledge of finding the area of a quadrilateral by decomposing it to solving mathematical and real-word problems involving the area of a quadrilateral and composite figures. Students decompose composite figures to find the area. Students are then guided to use the area of a composite figure to solve for missing dimension lengths on the figure. Finally, students complete a self-evaluation to explain what they still do not understand and to ask questions about the lesson.			
 Component Summary	7.6.1 Warm-Up (~5 min.)	Students create a triangle on a coordinate plane closest to six square units (<i>SE p. 326</i>)	7.6.4 Collaboration (10 min.)	Collaboration to find missing math facts in equations and missing dimensions in rectangles and triangles (<i>SE p. 329</i>)
	7.6.2 Guided Instruction (10 min.)	Discover multiple ways to decompose a composite figure and find its area (<i>SE p. 327</i>)	7.6.5 Guided Practice (5-10 min.)	Guided practice finding missing dimension of a composite figure using the given measurements (<i>SE p. 329</i>)
	7.6.3 Guided Practice (10 min.)	Decompose composite figures into triangles and rectangles to find their area (<i>SE p. 328</i>)	7.6.6 Your Turn! (10 min.)	Individual practice finding the area of and missing dimensions in composite figures (<i>SE p. 330</i>)
	7.6.7 Wrap-Up (~5 min.)	Reflection on lesson (<i>SE p. 331</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Composite figure - Area of a composite figure <u>Additional Terms:</u> - Dimension - Composite figure - Area		- Four-Function calculator (Optional) Scratch, tracing, or graph paper (Optional) Scissors (Optional) Separate paper (Optional) Square and rectangle manipulatives	
				Additional Preparation Consider making copies of cutouts of each composite figure to allow students to manipulate the shape.

 Teacher Tips	Common Misconceptions - Students may think a figure can only be decomposed into two shapes when more than two shapes are needed to find the area. - Students may not realize they need to use two measurements to find a third measurement that is not given. - Students with gaps in spatial awareness may struggle with decomposing a composite figure. - When more than one side does not have a measurement already labeled, students may struggle with discovering which side they need to solve for.	Things to Consider - Spend time showing students that there are different ways to divide a composite figure. Encourage students to choose ways that have the most given side lengths. - Encourage students to decompose composite figures in different ways to find the best way to calculate the missing dimension. - Consider not giving students a four-function calculator for this lesson to build student's fluency on the operations and using rational values.
	Literacy and Language Routines Think-Pair-Share In 7.6.7 Wrap-Up, students engage in a reflection to think about and then discuss with a partner what they still have a question about in the lesson. This may help students get some of their questions answered by another student and help clarify concepts related to solving for missing dimensions on composite figures.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.3.1: Mathematical Fluency During 7.6.4 Collaboration, students apply their knowledge of math facts to find missing factors and area of rectangles and triangles to find missing dimensions on composite figures. Using background knowledge and discussing multiple approaches to solving for area or missing dimensions can position students to complete tasks accurately and with confidence.
	 Differentiate Instruction The intent of 7.6.2 Guided Instruction and 7.6.3 Guided Practice centers on using decomposition to find area, not finding area alone. Students who still struggle finding area should leverage On-Ramp to 6th Grade videos on finding Area. Consider having students preview this lesson by watching the On-Ramp video or have the video available to watch for students to apt-in during 7.6.6 Your Turn! questions. In the 7.6.5 Guided Practice portion of this lesson, students need to make sense of visual representations of composite figures to solve for the missing dimension. Consider providing visual and kinesthetic entry points to learning, including: - Graphing the given figures by hand on graph paper or using online tools like GeoGebra to get a better sense of how the side lengths relate to each other. - Providing graph paper for students to recreate the drawing to better understand how to decompose the shape. - Using tracing paper to trace, cut out, and manipulate composite figures; physically cutting paper to decompose the shapes and identify how to use given measurements to find missing dimensions. - Using graphic organizers or lists to keep track of steps for solving or different equations used to find area of smaller shapes within composite shape.	
Lesson Notes:		

7.6.1 Warm-Up: Closest to Six

Component Overview: Students create a triangle on a graph whose area is closest to where the digits one through nine are used only one time each as coordinates. Consider using Think-Pair-Share routines on this activity, as students explore the area of a triangle and how it is related to a rectangle.



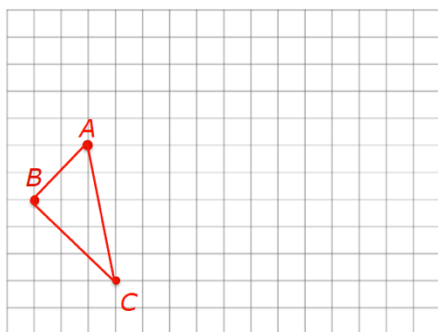
7.6.1 Warm-Up: Closest to Six

- Use the digits 0 to 9, at most one time each, to fill in ordered pairs for all three points, such that the area of Triangle ABC is closest to 6 square units.

A (3 , 6)

B (1 , 5)

C (4 , 2)



Think-Pair-Share

Consider using Think-Pair-Share routines as students test and determine an area of a triangle closest to six square units. As students share with partners, be sure to prompt students to think about what the possible side lengths are needed to get an area of 6. When students share ideas with a class, write responses on a shared classroom whiteboard so the whole class can see ideas and make connections or identify solutions that may not work.



As a visual approach for students struggling to get started, consider having students construct a rectangle whose area is equal to 12 square units using graph or tracing paper, then cutting it in half to represent 6 square units. Then have students move the triangle around the graph until they find points that are numerically different and cover approximately the same area.

7.6.2 Guided Instruction: Area of Composite Figures

Component Overview: Students will explore how to decompose a composite figure into multiple shapes and find their areas, whose sum will be the area of the composite figure. It is important to show students how there are multiple ways a shape can be decomposed and still get the correct area of the figure.

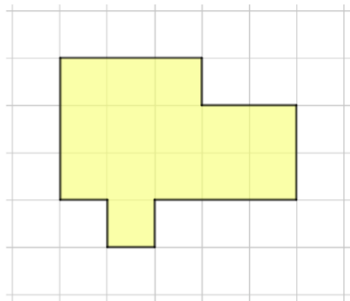


7.6.2 Guided Instruction: Area of Composite Figures



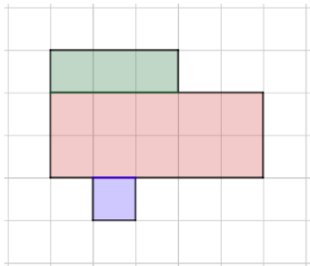
A **composite figure** is a two- or three-dimensional figure that can be decomposed into smaller figures.

Jackson and Danna were instructed to find the area of the composite figure below. The length of each square is 1 centimeter (cm).

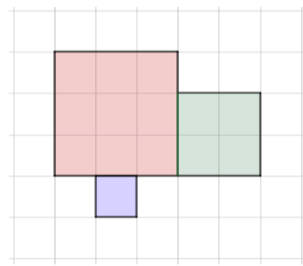


Both Jackson and Danna decomposed the figure, but their teacher noticed they each decomposed it differently.

Jackson



Danna



To highlight the multiple ways students can decompose composite figures, ask questions like:

- Is either student correct or incorrect? Why or why not?
- How are Jackson's and Danna's answers the same? How are they different?
- How would you have decomposed the composite shape?
- How could the decomposition help you find the area of the composite figure?

7.6.2 Guided Instruction: Area of Composite Figures (continued)

- Complete the table about each student's work. Include units.

Student	Area of Red Rectangle	Area of Blue Rectangle	Area of Green Rectangle
Jackson	10 cm^2	1 cm^2	3 cm^2
Danna	9 cm^2	1 cm^2	4 cm^2

- To find the area of the composite figure, find the sum of the areas of the rectangles for each student. Include units.

	Area of the Composite Figure
Jackson	14 cm^2
Danna	14 cm^2

- Explain why both students got the same area of the composite shape.

Both students got the same area of the composite shape because it does not matter which way you break down a shape, the area is the same.



The **area of a composite figure** is the sum of the areas of the figure's decomposed parts.



To further scaffold question 3, ask questions like:

- Does decomposing the composite figure change the shape overall? Why or why not?
- What do you notice about the areas of the smaller shapes? What does this mean for the area of the composite figure?

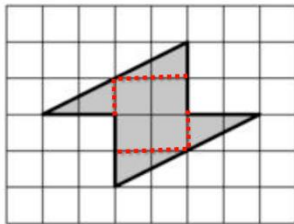
7.6.3 Guided Practice: Area of Composite Figures

Component Overview: Students will practice decomposing a composite figure. Students will be given a figure in which they can choose how to break up to find the areas of each shape in the figure to find the total area of the composite figure.



7.6.3 Guided Practice: Area of Composite Figures

1. A composite shape is shown. The length of each square is 1 inch (in).



- a. Draw lines to decompose the figure into rectangles and/or triangles.
- b. Find the area of each rectangle and/or triangle. Include units.

$1 \text{ in}^2, 1 \text{ in}^2, 1 \text{ in}^2, 1 \text{ in}^2, 4 \text{ in}^2$

- c. Find the sum of the areas of the rectangles and/or triangles to find the area of the composite figure. Include units.

8 in^2

- d. Find a classmate who decomposed the shape differently than you did. Describe their process.

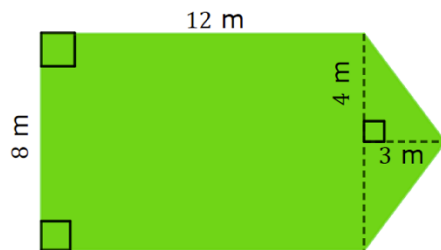
Answers will vary. My classmate divided the shape into two triangles. My classmate divided the shape into 3 triangles and 1 rectangle.



This activity provides visual entry points through the representations of the shapes shown. Auditory entry points can be provided through partner or whole-class discussion or reading aloud the questions. Consider providing cutouts of the shapes to provide kinesthetic entry points for all learners.

7.6.3 Guided Practice: Area of Composite Figures (continued)

2. Thomas has decided to put grass sod in his backyard. A drawing of his backyard is shown.



How much sod does Thomas need to buy?

$$A_{\text{rectangle}} = 8\text{ m} \times 12\text{ m} = 96\text{ m}^2$$

$$A_{\text{triangle}} = \frac{8\text{ m} \times 3\text{ m}}{2} = 12\text{ m}^2$$

$$A_{\text{total}} = A_{\text{rectangle}} + A_{\text{triangle}} = 96\text{ m}^2 + 12\text{ m}^2 = 108\text{ m}^2$$



Think-Pair-Share

Before answering question 2, engage students in a Think-Pair-Share routine. Ask guiding questions to provide further scaffolding like:

- How many shapes can you decompose this composite figure into?
- How can you use those shapes to find the area of the composite figure?

Have students first share their responses with a partner. Then share out as a class and clarify any misconceptions.



Consider prompting students to utilize a manipulative like a piece of paper for replicating this decomposition.

7.6.4 Collaboration: Finding a Missing Value

Component Overview: Students use operations reasoning to find a missing value in a multiplication or division equation, then determine how it relates to an equation using the same numbers in a different order with a different operation. Students then explore how to transfer this knowledge to finding the missing side length of a shape. Consider partnering students to engage in math discussions and check answers as they collaborate.



7.6.4 Collaboration: Finding a Missing Value

- Complete the table by filling in the blanks for the following math facts.

If.....	Then....
$40 \div 8 = \underline{5}$	$8 \times \underline{5} = 40$
$27 \div 3 \div \frac{1}{2} = \underline{18}$	$\frac{1}{2} \times \underline{18} \times 3 = 27$
$8 \div \frac{8}{7} = \underline{7}$	$\underline{7} \times \frac{8}{7} = 8$

- With your partner, use the previous math facts to find the missing dimensions of the quadrilaterals. All measurements are in centimeters (cm).

Area = 40 cm ²	Area = 27 cm ²	Area = 8 cm ²
$k = \underline{5}$	$m = \underline{18}$	$x = \underline{7}$



For question 1, encourage students to make connections between prior knowledge and the geometry tasks in this section by asking students:

- Which operations are inverses?
- How can inverse operations be used to find a missing number?



Use visual approaches to see the connection between the formula and diagrams shown by having students write the area formulas for each shape and substitute the given information until they are left with a variable to solve for.



For question 2, prompt student thinking about the relationship between the shapes' areas and missing dimensions by asking:

- What is the relationship between the area of a shape and the given dimension?
- How do you find the area of a triangle and rectangle?
- Where in the real world would you need to find a missing dimension?

7.6.5 Guided Practice: Finding the Value of the Missing Dimension

Component Overview: Students work individually to practice finding the missing dimension of a composite figure using given measurements. This activity combines prior knowledge of how to find a missing value in an equation with using an area equation to find the missing dimension.



7.6.5 Guided Practice: Finding the Value of the Missing Dimension

- Xavier is painting the back wall of a dollhouse. He knows the back wall of the dollhouse has an area of 6.875 square feet (ft^2).

- What is the area of the triangular top? Include units.

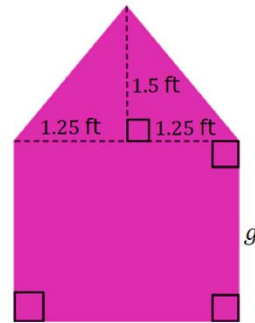
1.875 ft^2

- What is the area of the rectangular base? Include units.

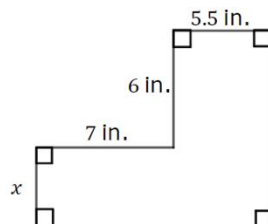
5 ft^2

- What is the height, g , of the rectangular base? Include units.

2 ft



- The area of the composite figure is 84 square inches (in^2). Find the missing dimension. Include units.



$$x = 4.08 \text{ in}^2$$



As students engage in question 1, provide questions to scaffold student thinking and reasoning about composite figures and finding area using properties of operations.

- How do you find the area of a triangle?
- Which side is the base of the triangle? How can you tell?
- What can you multiply by 1.25 to get the base of the triangle measurement? What's another expression you could use with addition?
- What is the formula for area of a triangle? What happens when you find $\frac{1}{2}$ of $2(1.25)$?
- How can you use this information to simplify your work before multiplying?



Students may incorrectly label sides using given measurements instead of solving for the unknown side lengths. Consider providing string as a measuring tool for students to verify measurements.



Contemplate, then Calculate

Prompt students to consider their approach to solving before calculating. Infusing this metacognitive strategy will contribute to fluency and mitigate errors.

7.6.6 Your Turn!

Component Overview: Students will demonstrate understanding of decomposing a composite figure to find the area of the shapes that make up that figure in order to find the total area using the given measurements.



7.6.6 Your Turn!

For each composite figure, find the area, in square feet (ft^2). Show your work.

1.

$$A_1 = 4 \text{ ft} \times (7.4 \text{ ft} - 5 \text{ ft}) = 9.6 \text{ ft}^2$$

$$A_2 = 5 \text{ ft} \times 2.8 \text{ ft} = 14 \text{ ft}^2$$

$$A_T = A_1 + A_2 = 9.6 \text{ ft}^2 + 14 \text{ ft}^2 = 23.6 \text{ ft}^2$$

Area = 23.6 ft^2



Consider differentiating by content with one of the following strategies:

- Choice: Allow students to choose two out of three questions to complete.
- Gridlines: Provide students with graph paper to trace each composite figure, then use the gridlines to better make sense of ways to decompose each figure.

2.

$$A_1 = \frac{(7.4 \text{ ft}) \times (2 \text{ ft})}{2} = 7.4 \text{ ft}^2$$

$$A_2 = 5.4 \text{ ft} \times 2.8 \text{ ft} = 15.12 \text{ ft}^2$$

$$A_T = 7.4 \text{ ft}^2 + 15.12 \text{ ft}^2 = 22.52 \text{ ft}^2$$

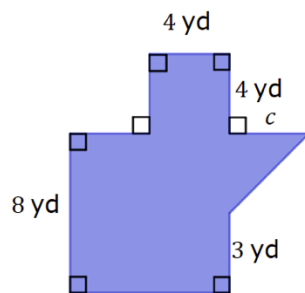
Area = 22.52 ft^2



Challenge students who finish early to create their own composite figure, including dimensions with at least one missing dimension, to trade with a partner and submit for extra credit.

7.6.6 Your Turn! (continued)

3. The area of the composite figure is 87.5 square yards (yd^2). Find the value of the missing dimension. Include units.



Not drawn to scale

$$c = 3 \text{ yd}$$



As a kinesthetic entry point, consider giving students a sheet of paper to draw the shapes that compose the composite figure. Have students cut out the shapes to visualize which measurement is missing. Also consider having students graph the given figures by hand or on GeoGebra to get a better sense of how the side lengths relate to each other.



For individual students who may have difficulty getting started, initiate geometric reasoning and problem-solving by asking:

- *How did you approach decomposing this figure? What were your reasons for decomposing it that way?*
- *How can you tell what the height of the triangle is? If you know the height and you are solving for the missing dimension c as the base, what other value do we need to solve for c ?*

7.6.7 Wrap-Up: Reflection

Component Overview: This Wrap-Up positions students to reflect on the lesson's learning targets and provide an opportunity to ask remaining questions about finding area of composite figures and/or finding missing dimensions. Consider using an auditory, collaborative approach to this reflection by having students share their questions with a partner and seek answers to clarify any misconceptions.



7.6.7 Wrap-Up: Reflection

Complete each statement based about today's lesson.

1. I am still unsure about ...
Answers will vary.
2. One question I still have about today's lesson is ...
Answers will vary.



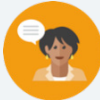
Think-Pair-Share

After students have had a few minutes to consider their responses, allow students to turn and talk to discuss with a partner what they still have a question about in the lesson. This may help students get some of their questions answered by another student and help clarify concepts for themselves or others.

7.7 Finding the Area of Polygons

Lesson Introduction

 Benchmark	MA.7.GR.1.2 Solve mathematical or real-world problems involving the area of polygons or composite figures by decomposing them into triangles or quadrilaterals.		 Learning Targets	- I can decompose a regular polygon into triangles to find the area. - I can decompose a regular polygon into triangles and quadrilaterals to find the area.	
 Benchmark Coherence	Building On MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.		Working Toward MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.		
 Essential Questions	- How do you decompose regular polygons into triangles and quadrilaterals to find the area? - How do you decompose regular polygons into triangles to find the area?				
 Lesson Narrative	Students derived the area formulas for trapezoids, parallelograms, and rhombi in previous lessons. In this lesson, students decompose regular polygons into only triangles or triangles and quadrilaterals to find the areas.				
 Component Summary	7.7.1 Warm-Up (~5 min.)	Justify which rectangle is more like a square (<i>SE p. 332</i>)	7.7.3 Collaboration (10 min.)	Collaboratively decompose regular pentagons and hexagons to find areas (<i>SE p. 335</i>)	
	7.7.2 Guided Instruction (15 min.)	Decompose regular hexagons to find areas (<i>SE p. 333</i>)	7.7.4 Your Turn! (10 min.)	Decompose regular polygons to find areas (<i>SE p. 336</i>)	
	7.7.5 Wrap-Up (~5 min.)	Identify and correct errors in a sample of student work finding area of a nonagon (<i>SE p. 337</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> Regular polygon <u>Additional Terms:</u> N/A		(Optional) Pattern blocks (Optional) Highlighters		Consider placing the pattern blocks needed to model the decomposition of each polygon into a labeled bag or container for each student to use throughout the lesson.

 Teacher Tips	Common Misconceptions - Students may incorrectly think that the area of regular polygons decomposed into smaller non-congruent triangles can be found by multiplying the area of one triangle by the number of triangles. - Students may not include the units with their answers or may not raise the units to the power of 2. - Students may round mid-step.	Things to Consider <i>Students using the Workbook may benefit from having multiple copies of each shape to spend less time erasing or drawing, and more time making mistakes and refining their approaches to decomposition.</i> - Consider allowing students to use a calculator for the lesson. In problems that require several steps students should not round until the final step. Rounding mid-step could significantly change the answer. If a place value is not given for rounding, then determine what place value you would like your students to round to. Many assessments have students round to three decimal values. Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common careless errors that many students make. - If time is an issue, then consider assigning 7.7.4 Your Turn! for homework.
	Literacy and Language Routines Stronger and Clearer 7.7.1 Warm-Up asks students to identify which given rectangle is more like a square. Ask students to write a justification for their selection and share it with a partner. Ask the partner to provide feedback. Next, give students time to revise their justifications based on the feedback that they received. Finally, repeat the sharing, feedback, and revising cycle.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations Students decompose regular polygons into only triangles or triangles and parallelograms to determine their areas. Students work with concrete representations of the area formula for the polygons. Through the activities, students are guided from concrete, to pictorial, to abstract representations as their understanding of the areas progresses.
	 Differentiate Instruction	Consider having On-Ramp to 6 th Grade videos available for students to use as additional scaffolding during the lesson or to use as a “preview” before the lesson to practice their fluency with finding area. - Finding Area of Figures Using Unit Squares - Finding Area of Rectilinear Shapes - Using the Area Formula for Rectangles
Lesson Notes:		

7.7.1 Warm-Up: Which One?

Component Overview: Students are provided with the dimensions for two different rectangles and asked to explain which is more like a square. The instructional routine “Stronger and Clearer Each Time” provides an auditory entry point to help students refine their justifications.



7.7.1 Warm-Up: Which One?

Two rectangles are given.

Rectangle A

13 units by 15 units

Rectangle B

2 units by 3 units

1. Explain which rectangle is more like a square.

Rectangle A

Answers will vary. Since all of the sides of a square are the same length, the side length divided by the side length will equal 1. Dividing the side lengths of Rectangle A will give an answer closer to 1 than dividing the sides of Rectangle B.



Stronger and Clearer Each Time

Ask students to write a justification for their selection and share it with a partner. Ask the partner to provide feedback. Next, give students time to revise their justification based on the feedback they received. Finally, repeat the sharing, feedback, and revising cycle.

7.7.2 Guided Instruction: Decomposing Regular Polygons

Component Overview: Students explore how to decompose a regular octagon into either congruent triangles, or triangles and quadrilaterals. Students compare the areas and discover that they are the same for both decomposition methods.

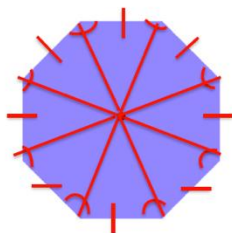


7.7.2 Guided Instruction: Decomposing Regular Polygons



A **regular polygon** is a polygon that is both equilateral (all sides congruent) and equiangular (all angles congruent).

1. A regular octagon is shown.



- a. On the figure, mark all congruent angles and all congruent sides.
See diagram above.
- b. Draw lines to decompose the regular octagon into congruent triangles.
See diagram above.
- c. How many congruent triangles did you decompose the regular octagon into?
8
- d. Discuss with your partner how decomposing the octagon into congruent triangles is helpful in determining the area.
Answers will vary. The area of the octagon will be the same as the sum of the area of all of the decomposed triangles.



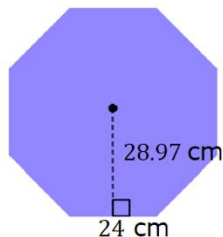
Students could decompose the octagon into 16 or 8 congruent triangles. Engage students in a discussion about which decomposition is correct and why. Prompt students to recognize that because all the triangles are congruent, they can multiply the area of one of the triangles by the number of triangles.



Consider providing students with pattern blocks to decompose the shapes throughout the lesson. This strategy could be replicated with paper and scissors.

7.7.2 Guided Instruction: Decomposing Regular Polygons (continued)

2. Some dimensions of an octagon are given in centimeters (cm). Each side of the octagon has a length of 24 cm. The perpendicular distance from the center to midpoint of a side is 28.97 cm.



- a. Find the area of one decomposed triangle. Round your answer to the nearest thousandth.

347.64 cm^2

- b. Find the area of the regular octagon rounded to the nearest thousandth of a square centimeter (cm^2) by finding the sum of the areas of all the decomposed triangles.

$2,781.12 \text{ cm}^2$



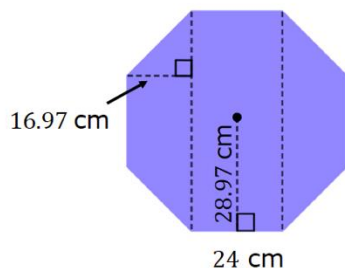
- Is this a regular octagon?
- How do you know?
- Would it be possible to find the area if this was not a regular octagon? Why or why not?



The answer to question 2a will depend on if students create 8 isosceles triangles or 16 right triangles. If a student decomposed the octagon into 16 right triangles, then the area of one decomposed triangle is 173.82 cm^2 . Notice all questions in this lesson ask for students to round to the nearest thousandth for consistency. However, the answer here only requires rounding to the hundredths place.

7.7.2 Guided Instruction: Decomposing Regular Polygons (continued)

3. An octagon is divided into two trapezoids and a rectangle. Each side of the octagon has a length of 24 cm. The perpendicular distance from the center to midpoint of a side is 28.97 cm. The height of each trapezoid is 16.97 cm.



- a. Find the area of each decomposed quadrilateral. Round to the nearest hundredth.
- b. Using the decomposed shapes, determine the area of the octagon rounded to the nearest thousandth of a square centimeter (cm^2).

695.26cm^2 ; 1390.56cm^2

$2,781.082\text{cm}^2$

4. Complete the chart using the areas from 2b and 3b

Decomposition	Area of Octagon
triangles	$2,781.12\text{cm}^2$
quadrilaterals	$2,781.08\text{cm}^2$

5. What do you notice about the areas?
Answers will vary. I notice that the areas are almost the same.



Consider providing students with pattern blocks to decompose the shapes throughout the lesson. This strategy could be replicated with paper and scissors.



Consider an inquiry-based approach when exploring question 3 with students.

- How did you find the lengths of the dimensions that were not given in the original drawing?
- Do you need to calculate the area of the trapezoid on the left and on the right?
- Would the areas be different when the octagon is decomposed into triangles versus when it is decomposed into triangles and quadrilaterals?



Help students understand how to put all the work for 3b into the Desmos calculator as

$$16.97(2 \cdot 28.97 + 24) + (2 \cdot 28.9) \cdot 24.$$

Explain why the $\frac{1}{2}$ is not necessary.

Students may wonder why the areas are slightly different. This occurs because the height of the trapezoid, 16.97, is a rounded value. This is a good opportunity to discuss which of these would be considered more accurate and how rounding can affect the outcome. Although this rounding was not something that the student did, it may be a good lesson on the dangers of rounding calculations along the way.

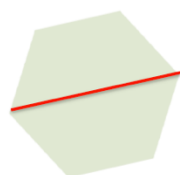
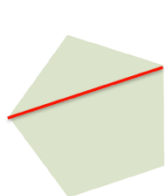
7.7.3 Collaboration: Decomposing Regular Polygons to Find Area

Component Overview: Students work with partners to decompose a regular pentagon and a regular hexagon into triangles and quadrilaterals to determine their areas. Question 1 should be completed individually by each student. Then, students work with partners for the remainder of the activity.



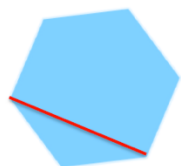
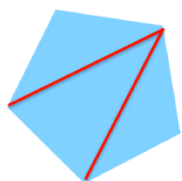
7.7.3 Collaboration: Decomposing Regular Polygons to Find Area

1. Decompose each of the following regular polygons.



Answers will vary.

2. Share with your partner how you decomposed your shapes. If your partner decomposed the shapes differently, draw on the blue shapes how they decomposed their shapes. If you decomposed the shapes in the same way, work together to create a different way to decompose the shapes.



Consider using question stems for students to guide their thinking while they collaborate or using these stems for pulling a small group during this activity.

Are there decompositions that would make it easier to find the area of the regular polygons?

If you decomposed the pentagon into triangles, are the triangles congruent?

If the triangles are not congruent, can you still use them to calculate area?

If you decomposed the hexagon into two trapezoids, are these trapezoids congruent? If you decomposed the hexagon into triangles, how many did you draw? Are they congruent?

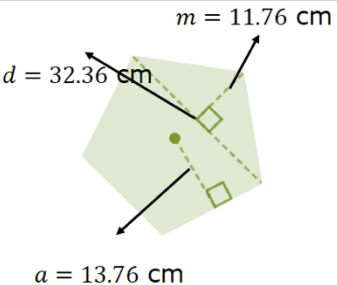
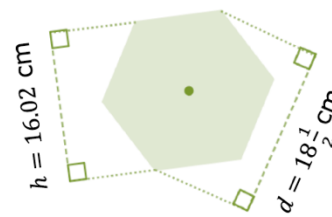
Choose one decomposition that you think would be easiest to use to find area. What lengths do you need to determine to find the areas?



If students decompose the pentagon into three triangles, they may incorrectly assume that the triangles are congruent. Prompt students to recognize that the three triangles are not congruent.

7.7.3 Collaboration: Decomposing Regular Polygons to Find Area (continued)

3. The table shows a pentagon and a hexagon. Given measurements in centimeters (cm). Work with your partner to determine the area rounded to the nearest hundredth.

Pentagon	Hexagon
 $m = 11.76$ cm $d = 32.36$ cm $a = 13.76$ cm Side lengths = 20 cm	 $h = 16.02$ cm $d = 18\frac{1}{2}$ cm Side lengths = $9\frac{1}{4}$ cm
Area of Pentagon 688 cm²	Area of Hexagon 222.28 cm²



Students with visual impairments or those struggling with decomposing shapes may be distracted by the many markings on the shapes in the table. Consider drawing a larger shape with dimensions for question 3 and using string or tape to label the dimensions.

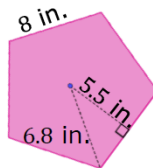
7.7.4 Your Turn!

Component Overview: Students practice independently decomposing regular polygons to find their areas. Students also analyze two samples of students' work. The students used different methods to decompose the regular hexagon. However, both methods result in the same area.



7.7.4 Your Turn!

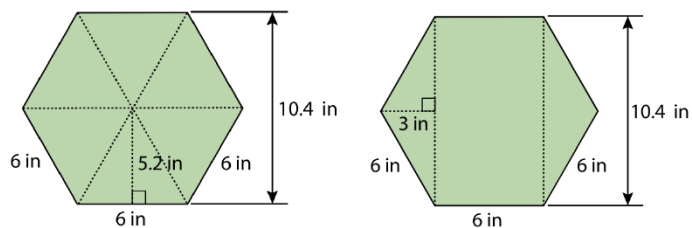
1. A regular pentagon is shown with lengths in inches (in.).



What is the area of the pentagon?

110 in^2

2. Lin and Andre used different methods to find the area of a regular hexagon with 6 in. sides. Lin decomposed the hexagon into six identical, equilateral triangles. Andre decomposed the hexagon into a rectangle and two triangles.



- a. Explain why Lin and Andre will find the same area of the hexagon.

Answers will vary. Lin and Andre will find the same area because decomposing the hexagon into different shapes does not change the area of the hexagon.



Students may misidentify the height of the triangle.



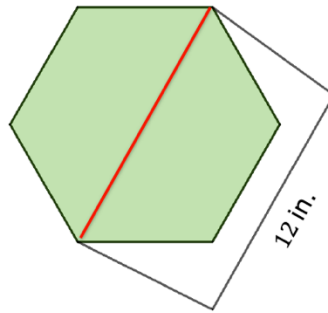
Consider having students outline the triangles and quadrilateral in different colors to distinguish between the shapes.

7.7.4 Your Turn! (continued)

- b. Find the area of the hexagon using both Lin's and Andre's methods.

Lin's Method	Andre's Method
93.6 in^2	93.6 in^2

- c. Decompose the shape differently than Lin and Andre did.



- d. Using the measurements given in all three figures, find the area by finding the sum of your decomposed parts.

$$93.6 \text{ in}^2$$



Prompt students to make a conjecture about the relationship between the number of sides in the regular polygon and the number of congruent triangles into which it can be decomposed. Ask students how this relationship can be used to create a general formula for the area of any regular polygon. Remind students to test their conjectures on a variety of different regular polygons.

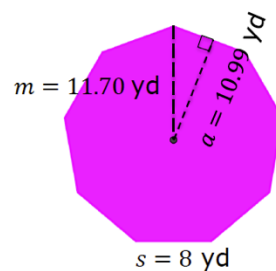
7.7.5 Wrap-Up: Error Analysis

Component Overview: Students identify and correct an error in a sample of a student's work. Consider using data from this formative assessment to determine which students need remediation before the next lesson.



7.7.5 Wrap-Up: Error Analysis

- Vernon found the area of the nonagon. His work is shown.



Students with visual impairments may struggle to read the dimensions on the nonagon or Vernon's work. Consider typing Vernon's work and enlarging the diagram of the nonagon.

Vernon's Work

Area of Triangle	Area of Nonagon
$A = \frac{1}{2}(8)(11.70)$	$A = 18(46.8)$
$A = 46.8 \text{ yd}^2$	$A = 842.4 \text{ yd}^2$

- Explain to Vernon what error(s) he made in his work.

Answers will vary. Vernon made the errors of using the wrong height and multiplying the area by 18.

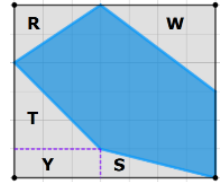
- What is the area of the nonagon?

395.64 yd^2

7.8 Finding the Area of Composite Figures

Lesson Introduction

 Benchmark	MA.7.GR.1.2 Solve mathematical or real-world problems involving the areas of polygons or composite figures by decomposing them into triangles or quadrilaterals.		 Learning Target	I can decompose composite figures into triangles and quadrilaterals to find the area.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.		MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.	
 Essential Questions	- How do you decompose composite figures into triangles and quadrilaterals to find the area? - How do you find the area of a composite figure inscribed in a rectangle?			
 Lesson Narrative	Students explore the similarities and differences between decomposing a composite figure to find its area and inscribing that composite figure within a rectangle to find the area in a different way. Students interact with triangles and rectangles as they work to decompose composite figures while making meaning of the purpose and intent of inscribing within a rectangle.			
 Component Summary	7.8.1 Warm-Up (~5 min.)	Identify math errors in an advertisement (<i>SE p. 338</i>)	7.8.4 Guided Practice (10 min.)	Practice decomposing composite figures to find their areas (<i>SE p. 341</i>)
	7.8.2 Guided Instruction (20 min.)	Find areas of composite figures by decomposing and by inscribing them in rectangles (<i>SE p. 339</i>)	7.8.5 Your Turn! (5 min.)	Find the area of a composite polygon (<i>SE p. 342</i>)
	7.8.3 Collaboration (10 min.)	Collaboratively decomposing a composite figure on grid paper to find its area (<i>SE p. 340</i>)	7.8.6 Wrap-Up (~5 min.)	Reflect on mastery of lesson learning targets (<i>SE p. 343</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> Composite figure <u>Additional Terms:</u> N/A		(Optional) Graph paper (Optional) Scissors (Optional) Markers	
			Additional Preparation	
			N/A	

	Common Misconceptions		Things to Consider	
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard	
 Teacher Tips	After decomposing the composite figures, students may struggle to find the lengths of sides not given in the triangles or quadrilaterals.		- In 7.8.4 Guided Practice prompt students to create a table with the types of shapes they decompose the figures into. Creating a table (like the one in 7.8.3 Collaboration) will help students organize their work. - Consider allowing students to use calculators. - If time is an issue, then consider assigning 7.8.4 Guided Practice and/or 7.8.5 Your Turn! for homework.	
	Think-Pair-Share During 7.8.1 Warm-Up prompt students to initially think about the math fail and write their response independently. Next, prompt students to share their thinking with a partner. Partners can provide each other with feedback before the whole-class discussion.		MA.K12.MTR.4.1: Discussion and Reflection Students collaborate with a partner in 7.8.3 Collaboration as they find the areas of composite figures. Students have opportunities to discuss their thinking with their peers. Select and highlight student work that advances and deepens the development of correct and increasingly efficient methods.	
 Differentiate Instruction	In 7.8.3 Collaboration question 1, students can use a graphic organizer to keep track of similarities and differences, with the potential to use different colors to show their partner's method of decomposing the figure. In 7.8.2 Guided Instruction consider prompting students to use graph paper to draw the figure and inscribe it in a rectangle. Students can cut out each shape that is not a part of the composite shape. Then, students can calculate and record the area for each one.			

Lesson Notes:

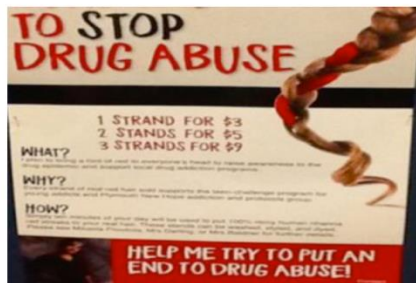
7.8.1 Warm-Up: Math Fail

Component Overview: Students identify and explain mistakes in a photograph of a sign. Use the Think-Pair-Share routine to engage student thinking and interaction.



7.8.1 Warm-Up: Math Fail

The advertisement below is for the sale of colored strands for braids to raise money for a Stop Drug Abuse campaign.



1. Explain what the math fail is in the advertisement.
Answers will vary. The math fail is that the rates are inconsistent; the 5 to 2 rate is not the same as the other two 3 to 1 rates.



Encourage creativity and thinking outside of the box. Students could respond in different ways, and both be 'correct' depending on their justification.

- Students may identify a 3:1 ratio, stating that two threads should cost \$6.
- Students may believe that unit price should decrease as quantity increases to encourage people to buy more. So, if buying 2 for \$5 saves a dollar, then buying 3 could cost \$7 to save \$2.

7.8.2 Guided Instruction: Finding the Area of Polygons

Component Overview: Students decompose composite figures into triangles and quadrilaterals. Then, they determine the dimensions of the triangles and quadrilaterals and find their areas. Next, students calculate the sum of the areas to determine the area of the composite figure. Students also inscribe a composite figure in a rectangle. They will calculate the areas of triangles and quadrilaterals outside of the composite figure. Students subtract the sum of these areas from the area of the rectangle to determine the area of the composite figure.

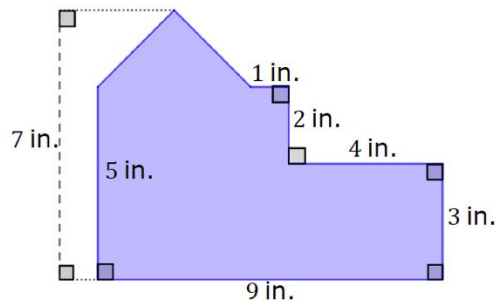


7.8.2 Guided Instruction: Finding the Area of Polygons



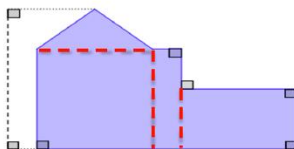
Composite figures are two- or three-dimensional figures that can be decomposed into smaller figures.

A floorplan is shown. All segment lengths are in inches (in.).



- With your partner, decompose the floorplan into triangles and quadrilaterals.

Answers will vary.



- Find the area of the floorplan. Show your work.

$$A = \frac{1}{2}bh; A = \frac{1}{2} \cdot 4 \cdot 2; A = \frac{1}{2} \cdot 8; A = 4 \text{ in}^2$$

$$A = bh; A = 4 \cdot 5; A = 20 \text{ in}^2$$

$$A = bh; A = 1 \cdot 5; A = 5 \text{ in}^2$$

$$A = bh; A = 4 \cdot 3; A = 12 \text{ in}^2$$

$$\text{Total Area: } 4 + 20 + 5 + 12 = 41 \text{ in}^2$$



What shapes did you decompose your figure into? How can you use the given dimensions to determine the unknown dimensions?

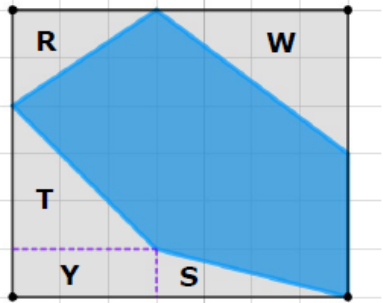


Think-Pair-Share

Prompt students to initially think about their mistakes and write their responses independently. Next, prompt students to share their thinking with a partner. Partners can provide each other with feedback before the whole-class discussion.

7.8.2 Guided Instruction: Finding the Area of Polygons (continued)

3. A polygon inscribed inside a rectangle is shown on the grid. Each grid line is 1 unit.
- a. Complete the table.

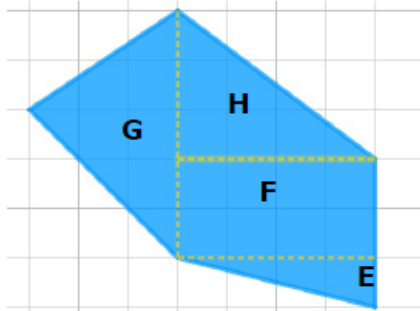
Polygon Inscribed in a Rectangle	
	
Decomposition of Rectangle	
Area of the large rectangle	42 unit^2
Area of R	3 unit^2
Area of S	2 unit^2
Area of T	4.5 unit^2
Area of W	6 unit^2
Area of Y	3 unit^2
Area of the blue polygon	23.5 unit^2



Consider having students use graph paper to draw and cut out a 6 by 7 unit rectangle. Then, students can draw and cut out R, S, T, Y, and W. This will provide a visual and kinesthetic entry-point for calculating the areas in the decomposition of the rectangle.

7.8.2 Guided Instruction: Finding the Area of Polygons (continued)

- b. The same polygon is shown on the grid. Each grid line is 1 unit.

Polygon

Decomposition of Polygon
Area of E 2 unit^2
Area of F 8 unit^2
Area of G 7.5 unit^2
Area of H 6 unit^2
Area of the blue polygon 23.5 unit^2

- c. Discuss with your partner the different possible methods of finding the area of the blue polygon. List one strength of each method.

Answers will vary. One method is to subtract the area surrounding the blue polygon from the area of the large rectangle. A strength is using shapes already defined. Another method is to decompose the blue polygon into various shapes and to find the total area of the shapes. A strength is finding the total area of less shapes.



What are the advantages of inscribing a composite figure into a rectangle to find area? What are the disadvantages? Which method do you prefer to use?



Think-Pair-Share

Prompt students to initially think about their mistakes and to write their responses independently. Next, prompt students to share their thinking with a partner. Partners can share feedback with each other before the whole-class discussion.

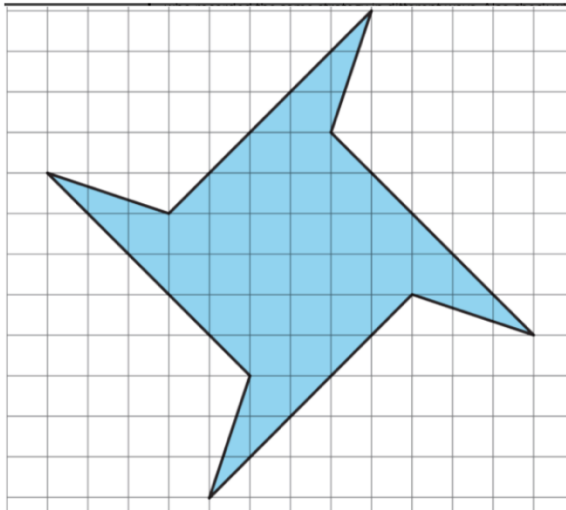
7.8.3 Collaboration: Area of a Polygon

Component Overview: Students collaborate to find various ways to decompose a composite figure, which is in the shape of a spinner. The students will find the area of each shape, and then find the total area of the composite figure.

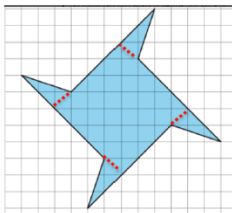


7.8.3 Collaboration: Area of a Polygon

1. A polygon is shown on grid paper. Each grid line is 1 unit.



- a. Decompose the polygon into triangles and quadrilaterals.



- b. Share your decomposition with your partner. List at least one similarity and one difference.

Answers will vary. Ours is similar because we decomposed it into the same amount of polygons. Ours is different because we decomposed it into a different amount of triangles and quadrilaterals.

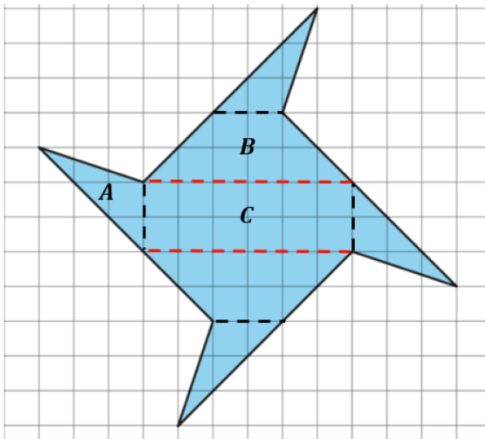


Provide colored pencils for students to sketch on their Workbook for support in decomposing, or provide copies of the shape on the coordinate plane for students to cut.

7.8.3 Collaboration: Area of a Polygon (continued)

2. Taahira decomposed the polygon into 7 pieces using 3 different shapes, as shown. Each grid line is 1 unit.

Work with your partner to find the area of each shape and the total area of the polygon.



Consider having students decompose to find the total area while also inscribing within a rectangle to verify their answers.

Students can work in pairs, with one student decomposing like Taahira while the partner is inscribing to verify. Students could also choose their own method with a partner before finding someone around the room who did it a different way to verify if they got the same answers.

Shape	Type of Shape	Area of the shape	Number of Shapes	Total Area
A	triangle	$3 u^2$	4	$12 u^2$
B	trapezoid	$8 u^2$	2	$16 u^2$
C	rectangle	$12 u^2$	1	12
Area of the Polygon				$40 u^2$

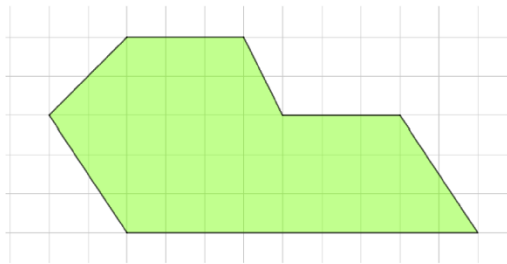
7.8.4 Guided Practice: Decomposing Composite Figures to Find Area

Component Overview: Students practice finding the areas of composite shapes. For question 1, students determine the dimensions by counting units on a grid. The second figure is not drawn on a grid, but the necessary dimensions are provided.



7.8.4 Guided Practice: Decomposing Composite Figures to Find Area

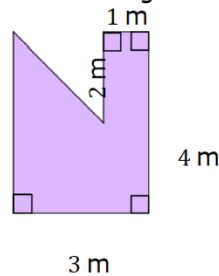
1. A polygon is shown on a grid. Each grid line is 1 unit.



Find the area of the polygon.

$36 u^2$

2. A polygon is shown with the given measurements in meters (m).



What is the area of the composite figure?

$10 m^2$



Encourage students to draw either:

- the decomposed triangles and quadrilaterals
- the rectangle in which the figure is inscribed

Prompt students to label the dimensions and make a table to organize the areas of the triangles and quadrilaterals.



What method did you use to determine the area? How can you use the given dimensions to determine the unknown dimensions?



Consider challenging students who finish early to find the areas of the two figures using a different approach. Prompt students to recognize that this is a way to check their work.

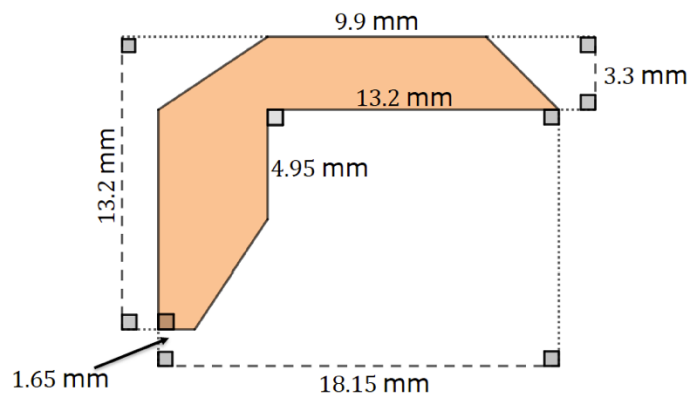
7.8.5 Your Turn!

Component Overview: Students practice independently finding the area of a composite figure. Consider providing students with graph paper and scissors so they can physically model the problem if they wish.



7.8.5 Your Turn!

- Find the area of the polygon. Lengths are shown in millimeters (mm).



87.12 mm^2



Consider challenging students who finish early to draw and label their own composite figures. Then, prompt students to swap figures with a partner and calculate the area of their partner's figure.

7.8.6 Wrap-Up: Reflection

Component Overview: Students reflect on mastery of lesson learning targets. Consider using data from this formative assessment to identify and address any student misunderstandings.



7.8.6 Wrap-Up: Reflection

1. Place an **X** on the number in the emoji scale that indicates your understanding in relation to today's learning target.

Answers will vary.

I can decompose composite figures into triangles and quadrilaterals to find the area.



2. Provide evidence to explain your self-assessment.

Answers will vary.



Consider creating a checklist designed for your learning targets. Students can create an emoji for each item or learning target.

I can decompose composite figures into triangles and quadrilaterals.

I can label the decomposed figures with the correct measures.

I can apply the correct formulas to find the area for each decomposed figure.

I can correctly calculate rational numbers to find the area.